

P25 J3361: Non-uniform Refinement

CryoSPARC Running Version: v4.7.1

Project Details

DETAIL	VALUE
Title	24sep20b_WT-Control-White_From-January
Description	This is a repeat collection of White Control Wildtype CFTR from the original January/February 2024 collection. Using
ID	P25
Created by user	Andy Paige

Job Details

DETAIL	VALUE
Title	New Job J3361
Description	Enter a description.
ID	J3361
Job type	Non-uniform Refinement
Status	completed
Created by user	Minou Emmad
Created	Fri Jun 05 2026 17:37:33 GMT-0400 (Eastern Daylight Time)
Queued	Fri Jun 05 2026 17:39:23 GMT-0400 (Eastern Daylight Time)
Launched	Fri Jun 05 2026 17:39:26 GMT-0400 (Eastern Daylight Time)
Started	Fri Jun 05 2026 17:39:34 GMT-0400 (Eastern Daylight Time)
Waiting	-
Killed	-
Completed	Fri Jun 05 2026 17:57:51 GMT-0400 (Eastern Daylight Time)
Failed	-
Last accessed	Wed Jun 10 2026 15:43:41 GMT-0400 (Eastern Daylight Time)
Size	5.27 GB

Inputs

particles

J3355.imported_particles

BLOB
J3355.imported_particles.blob.F

CTF
J3355.imported_particles.ctf.F

ALIGNMENTS3D

```
+-----+
| J3355.imported_particles.alignments3D.F |
+-----+
```

volume

```
J3358.volume_class_0
+-----+
| MAP |
+-----+
| J3358.volume_class_0.map.F |
+-----+
```

mask

No group connections

Parameters

Particle preprocessing

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Window dataset (real-space)	True	X		
Window inner radius	0.85	X		X
Window outer radius	0.99	X		X

Homogeneous Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Symmetry	C1	X		
Symmetry relaxation method	none	X		X
Do symmetry alignment	True	X		X
Re-estimate greyscale level of input reference	True	X		X
Number of extra final passes	0	X		X
Maximum align resolution (A)	None	X		
Initial lowpass resolution (A)	30	X		X
GSFSC split resolution (A)	20	X		X
Force re-do GS split	False	X		X
Adaptive Marginalization	True	X		X
Non-uniform refine enable	True	X		X
Non-uniform filter order	8	X		X
Non-uniform AWF	3	X		X
Enforce non-negativity	False	X		X
Window structure in real space	True	X		X
Skip interpolant premult	True	X		X
Ignore DC component	True	X		X
Ignore tilt	False	X		X
Ignore trefoil	False	X		X
Ignore tetra	False	X		X
Use random ordering of particles	True	X		X

	Disable auto batchsize		False		X				X	
	Batchsize epsilon		0.001		X				X	
	Batchsize snrfactor		50		X				X	
	Reset input per-particle scale		True		X				X	
	Minimize over per-particle scale		False		X				X	
	Scale min/use start iter		0		X				X	
	Noise priorw		50		X				X	
	Noise initw		200		X				X	
	Noise initial sigma-scale		3		X				X	
	Initialize noise model from images		False		X				X	
	Dynamic mask threshold (0-1)		0.2		X				X	
	Dynamic mask near (A)		6		X				X	
	Dynamic mask far (A)		14		X				X	
	Dynamic mask start resolution (A)		12		X					
	Dynamic mask use absolute value		False		X				X	
	Show plots from intermediate steps		True		X					
	GPU batch size of images		None		X				X	

Defocus Refinement

	PARAMETER		VALUE		DEFAULT		SPEC		ADVANCED	
	Optimize per-particle defocus		False		X				X	
	Num. particles to plot		3		X					
	Minimum Fit Res (A)		20		X					
	Defocus Search Range (A +/-)		2000		X					
	GPU batch size of images		None		X				X	

Global CTF Refinement

	PARAMETER		VALUE		DEFAULT		SPEC		ADVANCED	
	Optimize per-group CTF params		False		X				X	
	Num. groups to plot		3		X					
	Binning to apply to plots		1		X				X	
	Minimum Fit Res (A)		10		X					
	Fit Tilt		True		X					
	Fit Trefoil		True		X					
	Fit Spherical Aberration		False		X					
	Fit Tetrafoil		False		X					
	Fit Anisotropic Mag.		False		X					

GPU batch size of images	None	X		X	

Ewald Sphere Correction

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Do EWS correction	False	X		X	
EWS curvature sign	negative	X		X	

Random Seeds

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Random seed	817980808	X		X	

Compute settings

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Cache particle images on SSD	True	X			
Low-Memory Mode	False	X			

Outputs

particles

ALIGNMENTS3D	
J3361.particle.alignments3D	

CTF	
J3361.particle.ctf	

BLOB (PASSTHROUGH)	
J3361.particle.blob	

volume

MAP	
J3361.volume.blob	

MAP_SHARP	
J3361.volume.blob	

MAP_HALF_A	
J3361.volume.blob	

MAP_HALF_B	
J3361.volume.blob	

MASK_REFINE	

J3361.volume.blob	
MASK_FSC	
J3361.volume.blob	
MASK_FSC_AUTO	
J3361.volume.blob	
PRECISION	
J3361.volume.blob	

mask

MASK_REFINE	
J3361.volume.blob	

Target

ganges.biology.columbia.edu (node)

```
> [2026-06-05 17:39:26.92 GMT-4] License is valid.
> [2026-06-05 17:39:26.92 GMT-4] Launching job on lane default target ganges.biology.columbia.edu ...
> [2026-06-05 17:39:26.96 GMT-4] Running job on master node hostname ganges.biology.columbia.edu
> [2026-06-05 17:39:34.60 GMT-4] [CPU: 78.9 MB]      [Avail: 314.75 GB] Job J3361 Started
> [2026-06-05 17:39:34.63 GMT-4] [CPU: 79.0 MB]      [Avail: 314.64 GB] Master running v4.7.1, worker running v4.7.1
> [2026-06-05 17:39:34.65 GMT-4] [CPU: 79.2 MB]      [Avail: 314.61 GB] Working in directory:
                                /mnt/nfs_HuntLabSynologyDS1621CryoEM/hCFTR/Movies/24sep20b/24
                                sep20b/rawdata/CS-24sep20b-wt-control-white-from-january/J336
                                1
> [2026-06-05 17:39:34.65 GMT-4] [CPU: 79.2 MB]      [Avail: 314.60 GB] Running on lane default
> [2026-06-05 17:39:34.66 GMT-4] [CPU: 79.2 MB]      [Avail: 314.60 GB] Resources allocated:
> [2026-06-05 17:39:34.66 GMT-4] [CPU: 79.2 MB]      [Avail: 314.59 GB]   Worker:  ganges.biology.columbia.edu
> [2026-06-05 17:39:34.66 GMT-4] [CPU: 79.2 MB]      [Avail: 314.58 GB]   CPU    :  [8, 13, 14, 15]
> [2026-06-05 17:39:34.67 GMT-4] [CPU: 79.2 MB]      [Avail: 314.57 GB]   GPU    :  [2]
> [2026-06-05 17:39:34.67 GMT-4] [CPU: 79.2 MB]      [Avail: 314.56 GB]   RAM    :  [8, 9, 10]
> [2026-06-05 17:39:34.67 GMT-4] [CPU: 79.2 MB]      [Avail: 314.56 GB]   SSD    :  True
> [2026-06-05 17:39:34.68 GMT-4] [CPU: 79.2 MB]      [Avail: 314.59 GB] -----
> [2026-06-05 17:39:34.68 GMT-4] [CPU: 79.2 MB]      [Avail: 314.59 GB] -
> [2026-06-05 17:39:34.68 GMT-4] [CPU: 79.2 MB]      [Avail: 314.59 GB] Importing job module for job type nonuniform_refine_new...
> [2026-06-05 17:39:49.78 GMT-4] [CPU: 273.7 MB]      [Avail: 310.42 GB] Job ready to run
> [2026-06-05 17:39:49.78 GMT-4] [CPU: 273.7 MB]      [Avail: 310.41 GB] *****
> [2026-06-05 17:39:50.05 GMT-4] [CPU: 293.4 MB]      [Avail: 310.14 GB] **
> [2026-06-05 17:39:50.06 GMT-4] [CPU: 293.4 MB]      [Avail: 310.22 GB] Using random seed of 817980808
> [2026-06-05 17:39:50.08 GMT-4] [CPU: 293.4 MB]      [Avail: 310.20 GB] Loading a ParticleStack with 13211 items...

                                SSD cache ACTIVE at
                                /scr/instance_ganges.biology.columbia.edu:39001 (200 GB
                                reserve) (3 TB quota)

                                Cache usage                                Amount

                                Total / Usable                            3.58 TiB / 2.73 TiB
                                Used / Free                              2.73 TiB / 58.20 MiB
                                Hits / Misses                            66.12 GiB / 0.00 B
                                Acquired / Required                      66.12 GiB / 0.00 B

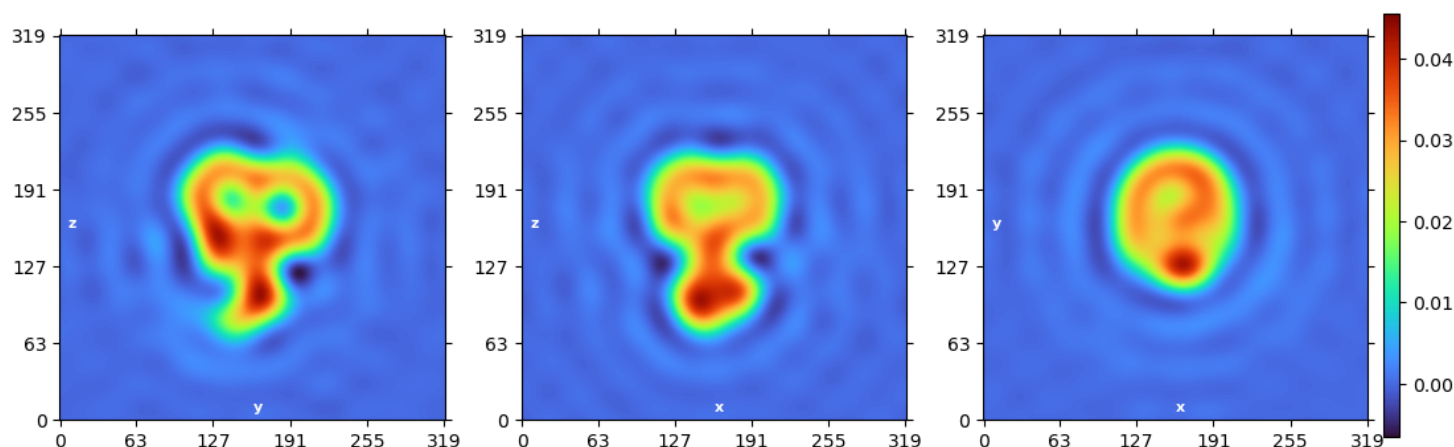
                                Progress:
                                [
                                6234/6234 (100%)
                                Elapsed: 0h 00m 06s
                                Active jobs: P31-J189, P31-J185, P25-J3361
                                SSD cache complete for 6234 file(s)

> [2026-06-05 17:39:57.72 GMT-4] [CPU: 336.7 MB]      [Avail: 314.11 GB] Done.
> [2026-06-05 17:39:57.72 GMT-4] [CPU: 336.7 MB]      [Avail: 314.11 GB] Windowing particles
> [2026-06-05 17:39:57.73 GMT-4] [CPU: 336.8 MB]      [Avail: 314.11 GB] Done.
> [2026-06-05 17:39:57.73 GMT-4] [CPU: 336.8 MB]      [Avail: 314.11 GB] ===== Gold Standard Split =====
> [2026-06-05 17:39:57.74 GMT-4] [CPU: 336.8 MB]      [Avail: 314.13 GB] Particles have input alignments3D connected, so reusing
                                pre-existing split
> [2026-06-05 17:39:57.74 GMT-4] [CPU: 336.8 MB]      [Avail: 314.16 GB] Set A is greater than set B by 95 particles (0.719 percent
                                difference relative to the total dataset).
> [2026-06-05 17:39:57.81 GMT-4] [CPU: 337.9 MB]      [Avail: 314.19 GB] Split A has 6653 particles
> [2026-06-05 17:39:57.81 GMT-4] [CPU: 337.9 MB]      [Avail: 314.19 GB] Split B has 6558 particles
> [2026-06-05 17:39:57.82 GMT-4] [CPU: 337.9 MB]      [Avail: 314.19 GB] ===== Refinement =====
> [2026-06-05 17:39:57.82 GMT-4] [CPU: 337.9 MB]      [Avail: 314.19 GB] Input particles have box size 320
> [2026-06-05 17:39:57.82 GMT-4] [CPU: 337.9 MB]      [Avail: 314.18 GB] Input particles have pixel size 0.8300
> [2026-06-05 17:39:57.82 GMT-4] [CPU: 337.9 MB]      [Avail: 314.18 GB] Particles will be zeropadded/truncated to size 320 during
                                alignment
```

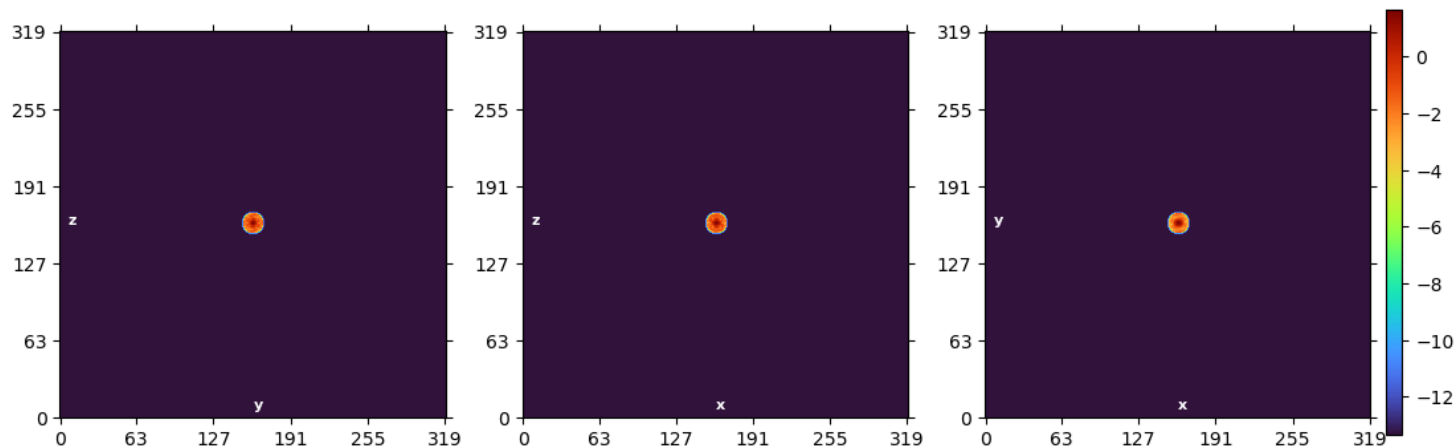
```

> [2026-06-05 17:39:57.83 GMT-4] [CPU: 337.9 MB] [Avail: 314.18 GB] Volume refinement will be done with effective box size 320
> [2026-06-05 17:39:57.83 GMT-4] [CPU: 337.9 MB] [Avail: 314.18 GB] Volume refinement will be done with pixel size 0.8300
> [2026-06-05 17:39:57.83 GMT-4] [CPU: 337.9 MB] [Avail: 314.18 GB] Particles will be zeropadded/truncated to size 320 during
backprojection
> [2026-06-05 17:39:57.84 GMT-4] [CPU: 337.9 MB] [Avail: 314.18 GB] Particles will be backprojected with box size 320
> [2026-06-05 17:39:57.84 GMT-4] [CPU: 338.1 MB] [Avail: 314.18 GB] Volume will be internally cropped and stored with box size
320
> [2026-06-05 17:39:57.84 GMT-4] [CPU: 338.1 MB] [Avail: 314.18 GB] Volume will be interpolated with box size 320 (zeropadding
factor 1.00)
> [2026-06-05 17:39:57.85 GMT-4] [CPU: 338.1 MB] [Avail: 314.17 GB] DC components of images will be ignored and volume will be
floated at each iteration.
> [2026-06-05 17:39:57.85 GMT-4] [CPU: 338.1 MB] [Avail: 314.17 GB] Spherical windowing of maps is enabled
> [2026-06-05 17:39:57.85 GMT-4] [CPU: 338.1 MB] [Avail: 314.17 GB] Refining with C1 symmetry enforced
> [2026-06-05 17:39:57.86 GMT-4] [CPU: 338.1 MB] [Avail: 314.17 GB] Refining with pose and shift marginalization during
backprojection enabled.
> [2026-06-05 17:39:57.86 GMT-4] [CPU: 338.1 MB] [Avail: 314.17 GB] Resetting input per-particle scale factors to 1.0
> [2026-06-05 17:39:57.86 GMT-4] [CPU: 338.1 MB] [Avail: 314.17 GB] Starting at initial resolution 30.000A (radwn 8.853).
> [2026-06-05 17:39:57.87 GMT-4] [CPU: 338.1 MB] [Avail: 314.17 GB] ===== Non-Uniform Refinement =====
> [2026-06-05 17:39:57.87 GMT-4] [CPU: 338.1 MB] [Avail: 314.16 GB] Non-Uniform Refinement is enabled.
> [2026-06-05 17:39:57.87 GMT-4] [CPU: 338.1 MB] [Avail: 314.16 GB] Using AWF of 3.00.
> [2026-06-05 17:39:57.88 GMT-4] [CPU: 338.1 MB] [Avail: 314.16 GB] Using butterworth filter with order 8.
> [2026-06-05 17:39:57.88 GMT-4] [CPU: 338.1 MB] [Avail: 314.16 GB] ===== Masking =====
> [2026-06-05 17:40:00.37 GMT-4] [CPU: 870.3 MB] [Avail: 315.85 GB] No mask input was connected, so dynamic masking will be
enabled.
> [2026-06-05 17:40:00.37 GMT-4] [CPU: 870.3 MB] [Avail: 315.85 GB] Dynamic mask threshold: 0.2000
> [2026-06-05 17:40:00.37 GMT-4] [CPU: 870.3 MB] [Avail: 315.86 GB] Dynamic mask near (A): 6.00
> [2026-06-05 17:40:00.38 GMT-4] [CPU: 870.3 MB] [Avail: 315.86 GB] Dynamic mask far (A): 14.00
> [2026-06-05 17:40:00.38 GMT-4] [CPU: 870.3 MB] [Avail: 315.86 GB] ===== Initial Model =====
> [2026-06-05 17:40:00.38 GMT-4] [CPU: 870.3 MB] [Avail: 315.86 GB] Resampling initial model to specified volume representation
size and pixel-size...
> [2026-06-05 17:40:03.36 GMT-4] [CPU: 1.25 GB] [Avail: 312.86 GB] Estimating scale of initial reference.
> [2026-06-05 17:40:07.92 GMT-4] [CPU: 77.4 MB] [Avail: 310.95 GB] WARNING: io_uring support disabled (not supported by kernel),
I/O performance may degrade
> [2026-06-05 17:40:09.78 GMT-4] [CPU: 1.46 GB] [Avail: 312.42 GB] Rescaling initial reference by a factor of 1.396
> [2026-06-05 17:40:11.41 GMT-4] [CPU: 1.46 GB] [Avail: 312.29 GB] Estimating scale of initial reference.
> [2026-06-05 17:40:14.08 GMT-4] [CPU: 1.42 GB] [Avail: 312.15 GB] Rescaling initial reference by a factor of 1.004
> [2026-06-05 17:40:15.76 GMT-4] [CPU: 1.42 GB] [Avail: 312.51 GB] Estimating scale of initial reference.
> [2026-06-05 17:40:18.51 GMT-4] [CPU: 1.42 GB] [Avail: 312.41 GB] Rescaling initial reference by a factor of 0.990
> [2026-06-05 17:40:21.13 GMT-4] Initial Real Space Slices

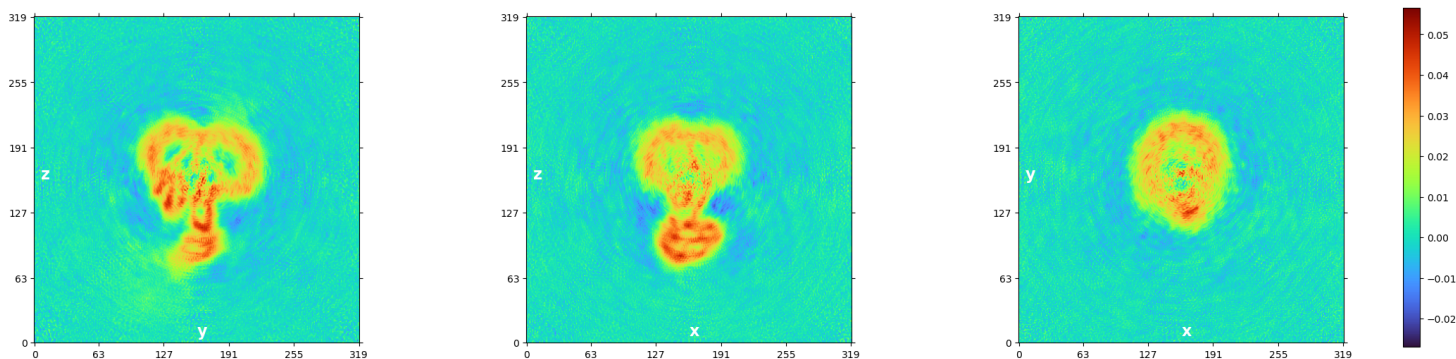
```



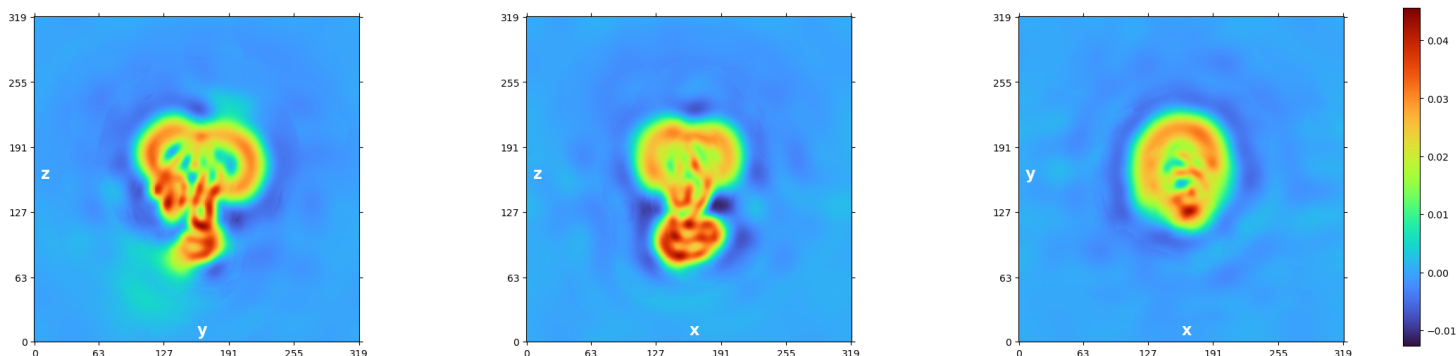
> [2026-06-05 17:40:22.55 GMT-4] Initial Fourier Space Slices



```
> [2026-06-05 17:40:22.55 GMT-4] [CPU: 1.47 GB] [Avail: 311.77 GB] ===== Starting Refinement Iterations =====
> [2026-06-05 17:40:22.56 GMT-4] [CPU: 1.47 GB] [Avail: 311.77 GB] ----- Start Iteration 0
> [2026-06-05 17:40:22.56 GMT-4] [CPU: 1.47 GB] [Avail: 311.77 GB] Using Max Alignment Radius 8.853 (30.000A)
> [2026-06-05 17:40:22.56 GMT-4] [CPU: 1.47 GB] [Avail: 311.77 GB] Auto batchsize: 5942 in each split
> [2026-06-05 17:40:26.19 GMT-4] [CPU: 1.98 GB] [Avail: 310.78 GB] -- THR 1 BATCH 500 NUM 1471 TOTAL 2.6516695 ELAPSED 31.434437
--
> [2026-06-05 17:40:59.85 GMT-4] [CPU: 3.13 GB] [Avail: 306.58 GB] Processed 11884.000 images in 33.657s.
> [2026-06-05 17:41:03.93 GMT-4] [CPU: 3.45 GB] [Avail: 306.81 GB] Computing FSCs...
> [2026-06-05 17:41:03.94 GMT-4] [CPU: 3.45 GB] [Avail: 306.82 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 17:41:03.94 GMT-4] [CPU: 3.45 GB] [Avail: 306.84 GB] Computing FFTs on GPU.
> [2026-06-05 17:41:06.53 GMT-4] [CPU: 3.70 GB] [Avail: 306.10 GB] Done in 2.593s
> [2026-06-05 17:41:06.53 GMT-4] [CPU: 3.70 GB] [Avail: 306.10 GB] Computing cFSCs...
> [2026-06-05 17:41:08.70 GMT-4] [CPU: 3.70 GB] [Avail: 305.89 GB] Done in 2.159s
> [2026-06-05 17:41:08.70 GMT-4] [CPU: 3.70 GB] [Avail: 305.89 GB] Using Filter Radius 30.252 (8.779A) | Previous: 8.853
(30.000A)
> [2026-06-05 17:41:14.88 GMT-4] [CPU: 4.77 GB] [Avail: 304.81 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 17:41:14.88 GMT-4] [CPU: 4.77 GB] [Avail: 304.80 GB] Running local cross validation for A ...
> [2026-06-05 17:41:43.70 GMT-4] [CPU: 7.44 GB] [Avail: 307.01 GB] Local cross validation A done in 28.810s
> [2026-06-05 17:41:45.36 GMT-4] FSC Filtered Side A
```

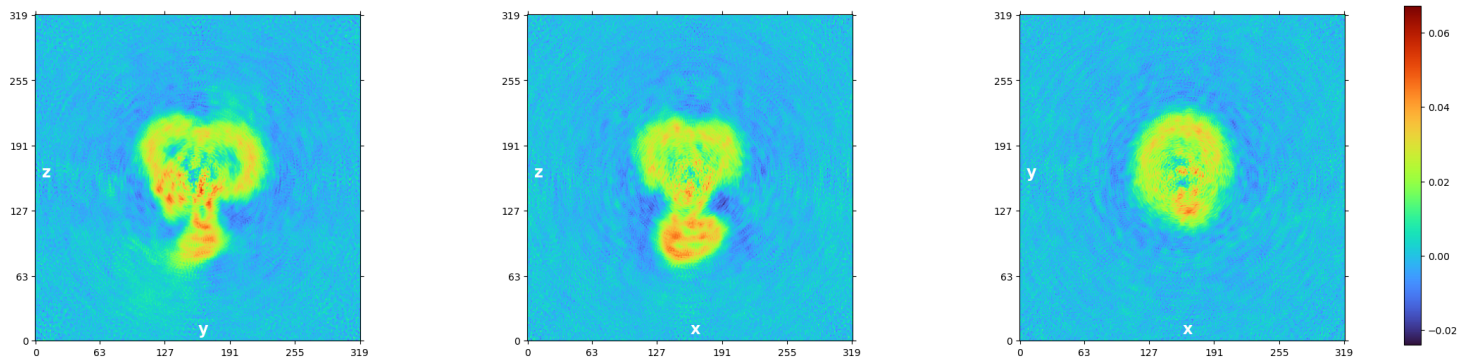


> [2026-06-05 17:41:46.63 GMT-4] CV Filtered Side A

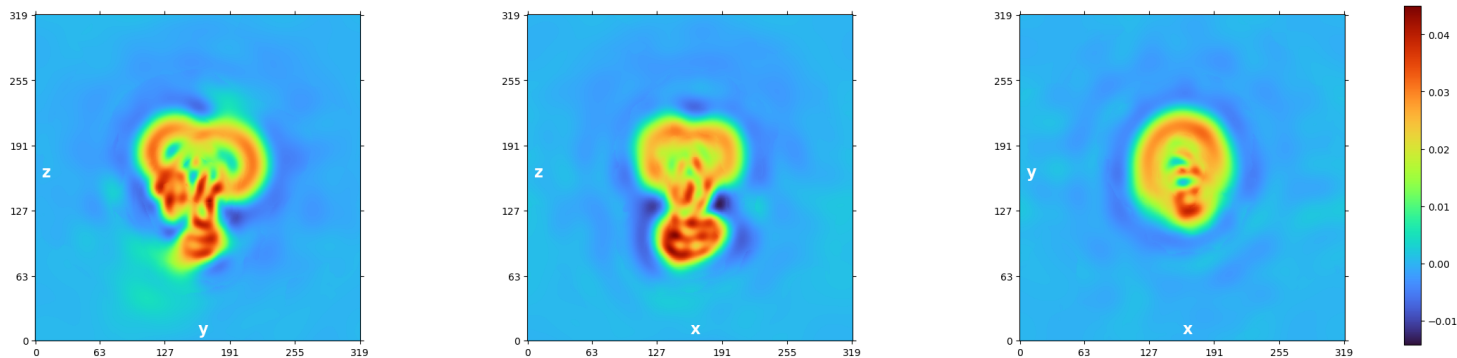


```
> [2026-06-05 17:41:46.63 GMT-4] [CPU: 7.45 GB] [Avail: 306.96 GB] Running local cross validation for B ...
> [2026-06-05 17:42:08.72 GMT-4] [CPU: 7.73 GB] [Avail: 305.70 GB] Local cross validation B done in 22.084s
```

> [2026-06-05 17:42:10.10 GMT-4] FSC Filtered Side B



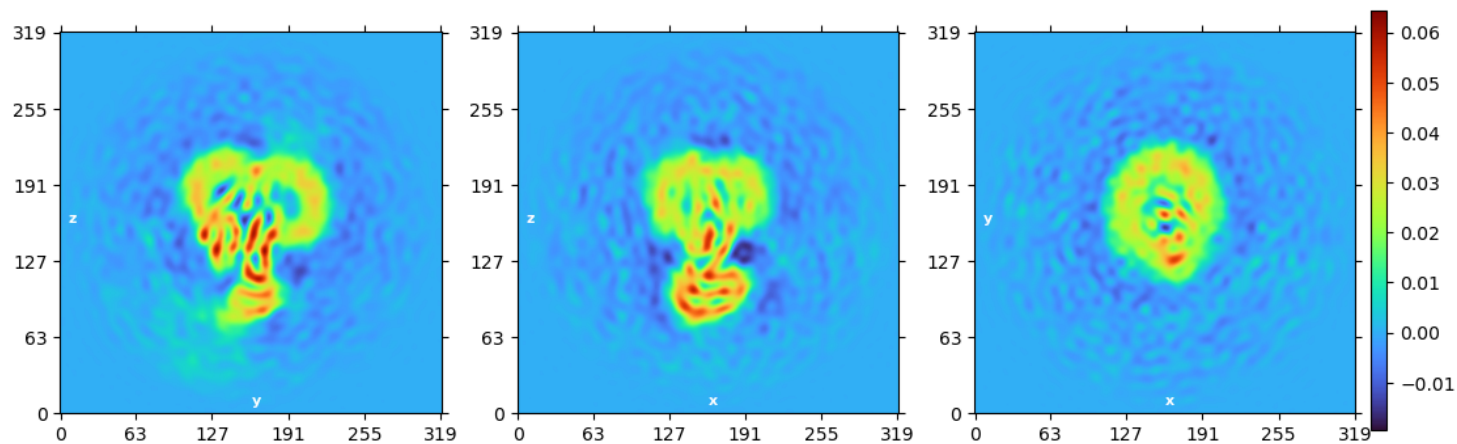
> [2026-06-05 17:42:11.34 GMT-4] CV Filtered Side B



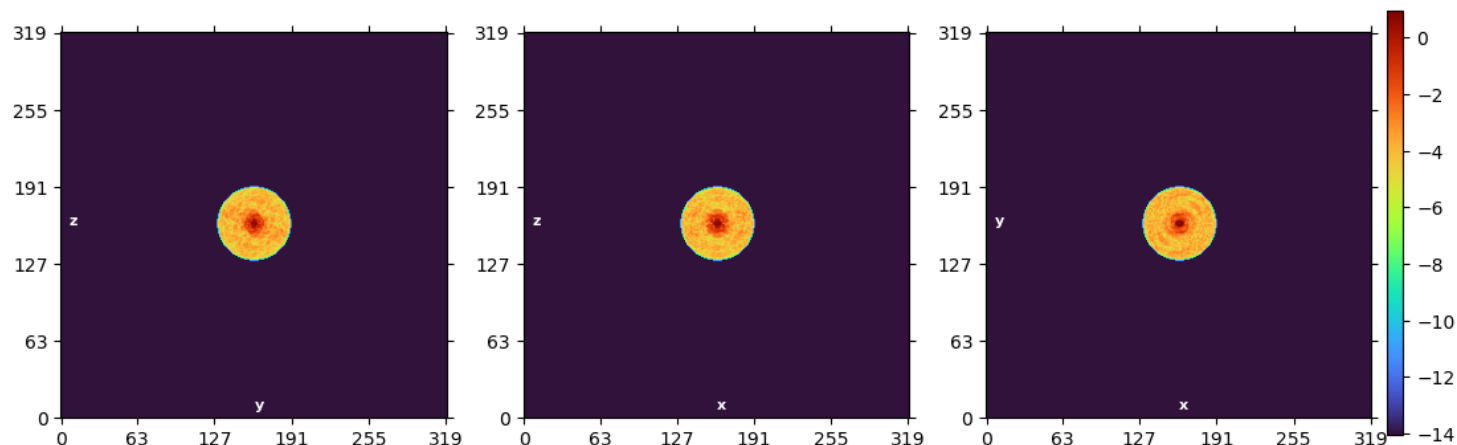
> [2026-06-05 17:42:19.41 GMT-4] [CPU: 8.06 GB] [Avail: 305.02 GB] Estimated Bfactor: -588.9

> [2026-06-05 17:42:19.42 GMT-4] [CPU: 8.06 GB] [Avail: 305.01 GB] Plotting..

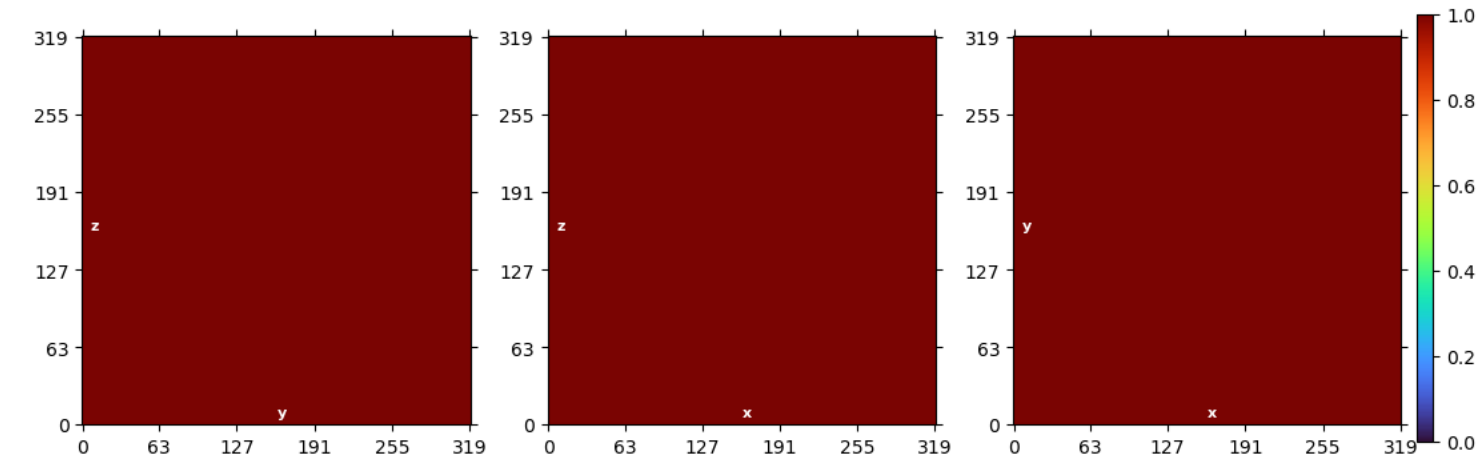
> [2026-06-05 17:42:26.10 GMT-4] Real Space Slices Iteration 000



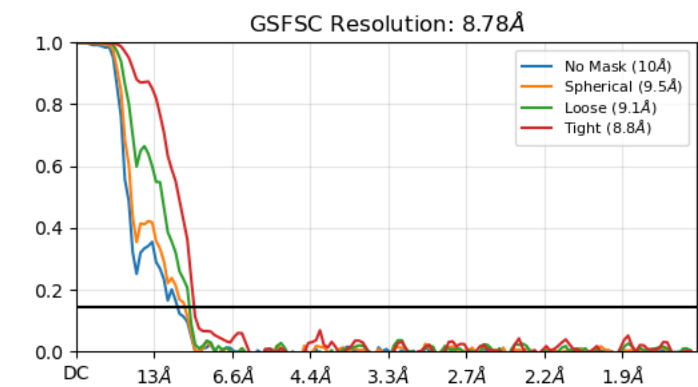
> [2026-06-05 17:42:27.58 GMT-4] Fourier Space Slices Iteration 000



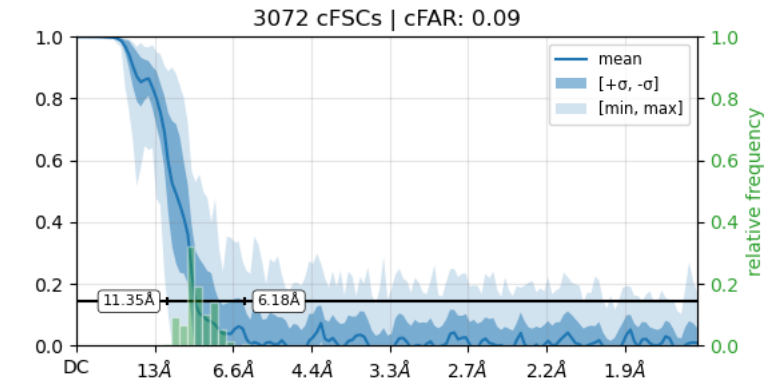
> [2026-06-05 17:42:28.83 GMT-4] Real Space Mask Slices Iteration 000



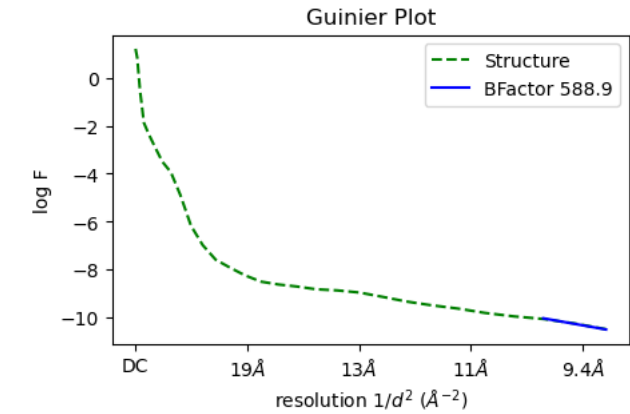
> [2026-06-05 17:42:29.60 GMT-4] FSC Iteration 000



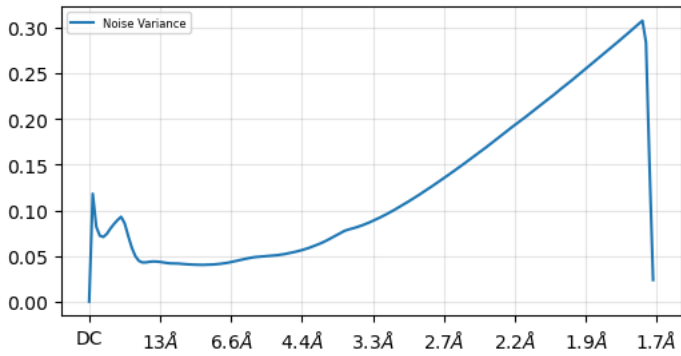
> [2026-06-05 17:42:30.89 GMT-4] cFSCs (Half-angle: 20°) Iteration 000, with tight mask



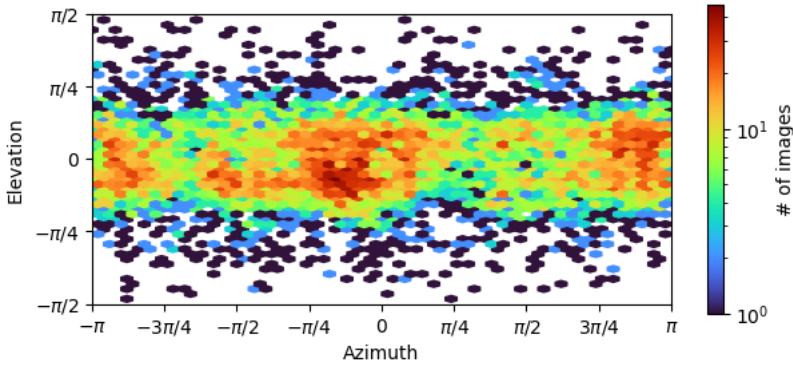
> [2026-06-05 17:42:36.60 GMT-4] Guinier Plot Iteration 000



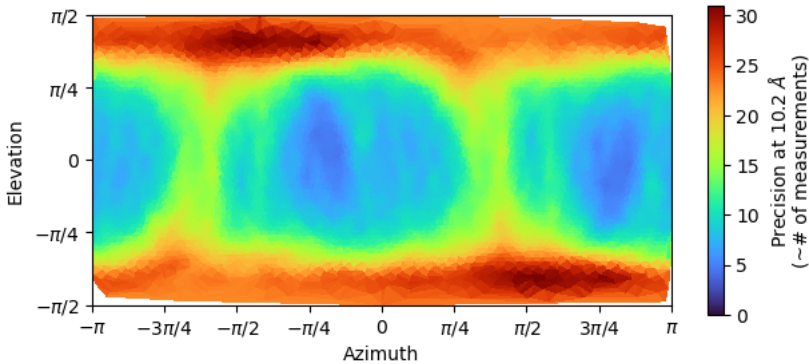
> [2026-06-05 17:42:37.01 GMT-4] Noise Model Iteration 000



> [2026-06-05 17:42:37.99 GMT-4] Viewing Direction Distribution Iteration 000

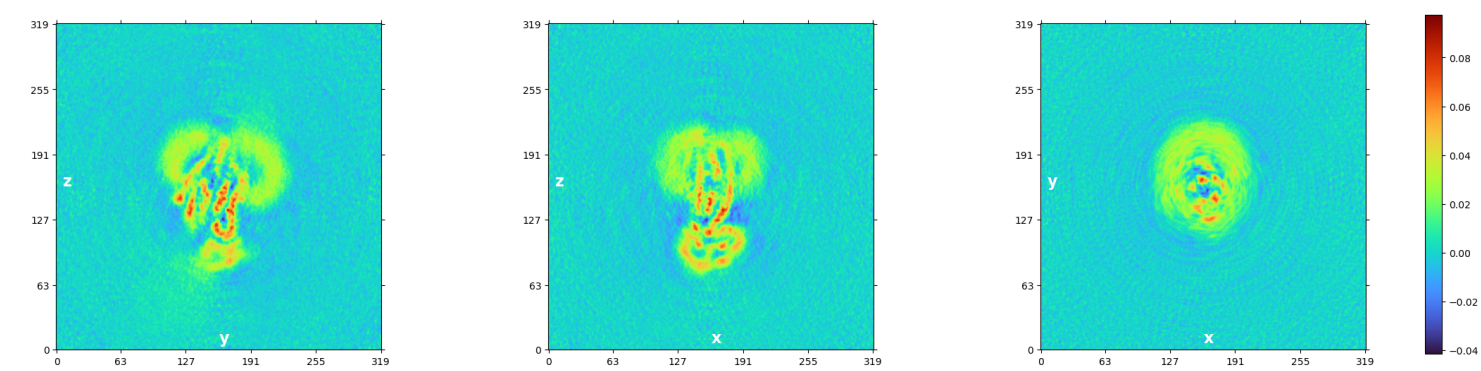


> [2026-06-05 17:42:39.69 GMT-4] Posterior Precision Directional Distribution Iteration 000

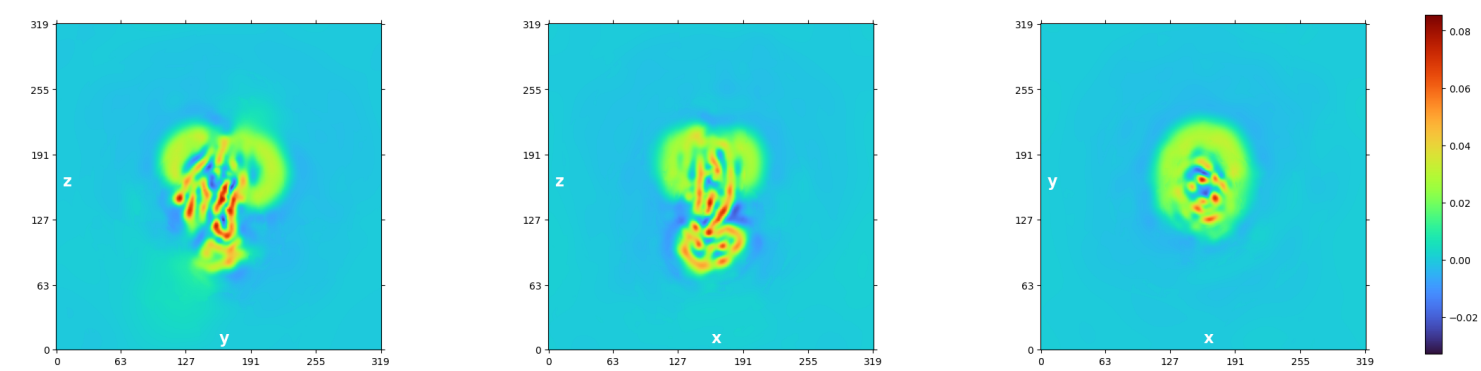


```
> [2026-06-05 17:42:39.69 GMT-4] [CPU: 8.48 GB] [Avail: 301.73 GB] Done in 20.273s.
> [2026-06-05 17:42:39.70 GMT-4] [CPU: 8.48 GB] [Avail: 301.73 GB] Outputting files..
> [2026-06-05 17:42:54.79 GMT-4] [CPU: 8.91 GB] [Avail: 302.50 GB] Done in 15.088s.
> [2026-06-05 17:42:54.79 GMT-4] [CPU: 8.91 GB] [Avail: 302.49 GB] Done iteration 0 in 152.231s. Total time so far 184.983s
> [2026-06-05 17:42:55.50 GMT-4] [CPU: 8.91 GB] [Avail: 301.20 GB] ----- Start Iteration 1
> [2026-06-05 17:42:55.51 GMT-4] [CPU: 8.91 GB] [Avail: 301.24 GB] Using Max Alignment Radius 30.252 (8.779A)
> [2026-06-05 17:42:55.51 GMT-4] [CPU: 8.91 GB] [Avail: 301.28 GB] Auto batchsize: 6653 in each split
> [2026-06-05 17:42:55.51 GMT-4] [CPU: 8.91 GB] [Avail: 301.30 GB] Using dynamic mask.
> [2026-06-05 17:43:25.65 GMT-4] [CPU: 9.41 GB] [Avail: 307.34 GB] -- THR 0 BATCH 500 NUM 1779 TOTAL 5.0941855 ELAPSED 32.413337
--
> [2026-06-05 17:44:00.56 GMT-4] [CPU: 10.28 GB] [Avail: 299.06 GB] Processed 13211.000 images in 34.914s.
> [2026-06-05 17:44:03.75 GMT-4] [CPU: 10.31 GB] [Avail: 298.72 GB] Computing FSCs...
> [2026-06-05 17:44:03.75 GMT-4] [CPU: 10.31 GB] [Avail: 298.72 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 17:44:03.75 GMT-4] [CPU: 10.31 GB] [Avail: 298.72 GB] Computing FFTs on GPU.
> [2026-06-05 17:44:05.76 GMT-4] [CPU: 10.31 GB] [Avail: 298.93 GB] Done in 2.013s
> [2026-06-05 17:44:05.77 GMT-4] [CPU: 10.31 GB] [Avail: 298.93 GB] Computing cFSCs...
> [2026-06-05 17:44:08.01 GMT-4] [CPU: 10.31 GB] [Avail: 299.43 GB] Done in 2.236s
> [2026-06-05 17:44:08.02 GMT-4] [CPU: 10.31 GB] [Avail: 299.44 GB] Using Filter Radius 60.277 (4.406A) | Previous: 30.252
(8.779A)
> [2026-06-05 17:44:14.80 GMT-4] [CPU: 9.56 GB] [Avail: 299.39 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 17:44:14.81 GMT-4] [CPU: 9.56 GB] [Avail: 299.39 GB] Running local cross validation for A ...
> [2026-06-05 17:44:37.15 GMT-4] [CPU: 9.78 GB] [Avail: 298.85 GB] Local cross validation A done in 22.340s
```

> [2026-06-05 17:44:38.65 GMT-4] FSC Filtered Side A



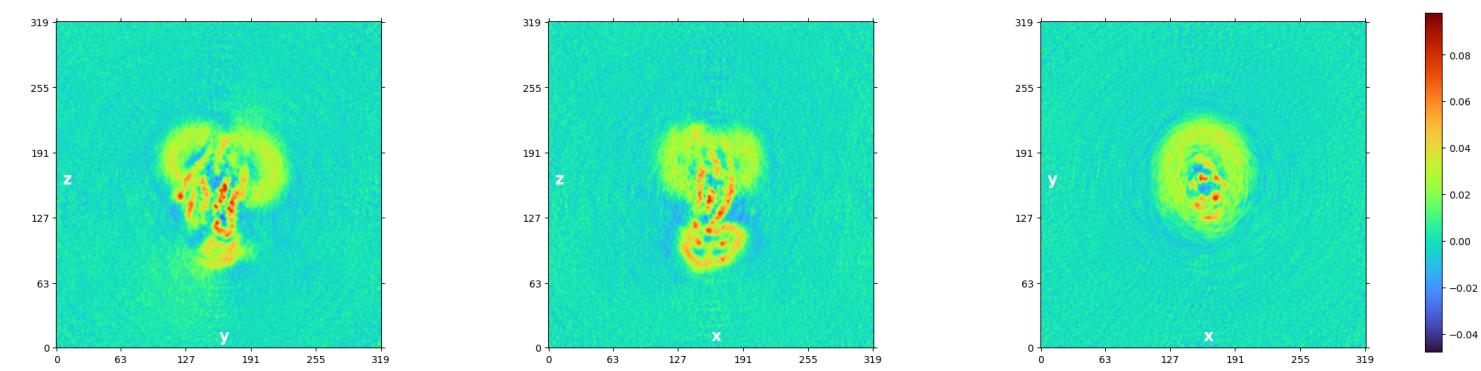
> [2026-06-05 17:44:39.78 GMT-4] CV Filtered Side A



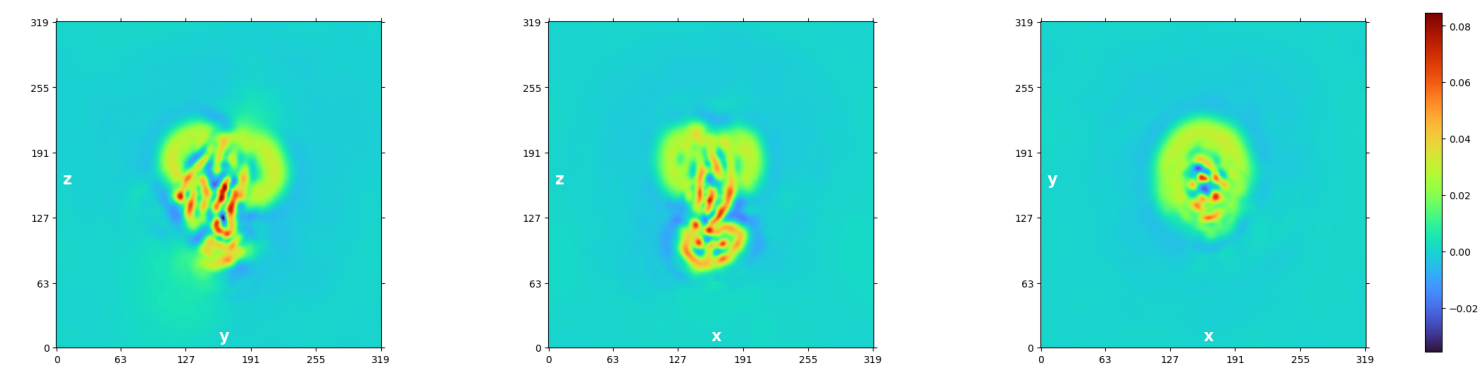
> [2026-06-05 17:44:39.78 GMT-4] [CPU: 9.78 GB] [Avail: 299.45 GB] Running local cross validation for B ...

> [2026-06-05 17:45:02.12 GMT-4] [CPU: 10.03 GB] [Avail: 307.60 GB] Local cross validation B done in 22.340s

> [2026-06-05 17:45:03.67 GMT-4] FSC Filtered Side B



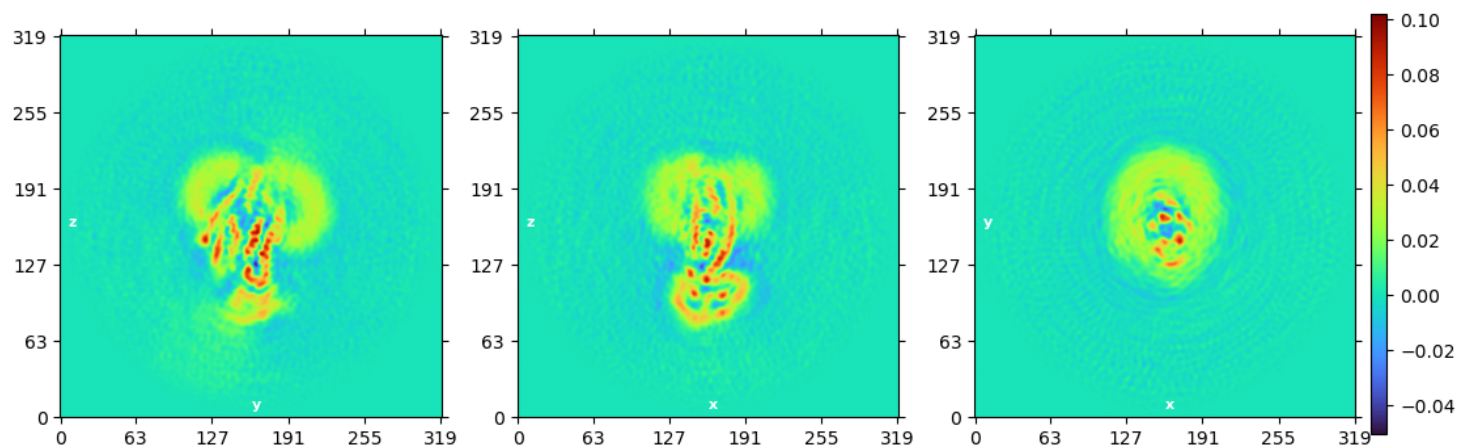
> [2026-06-05 17:45:04.88 GMT-4] CV Filtered Side B



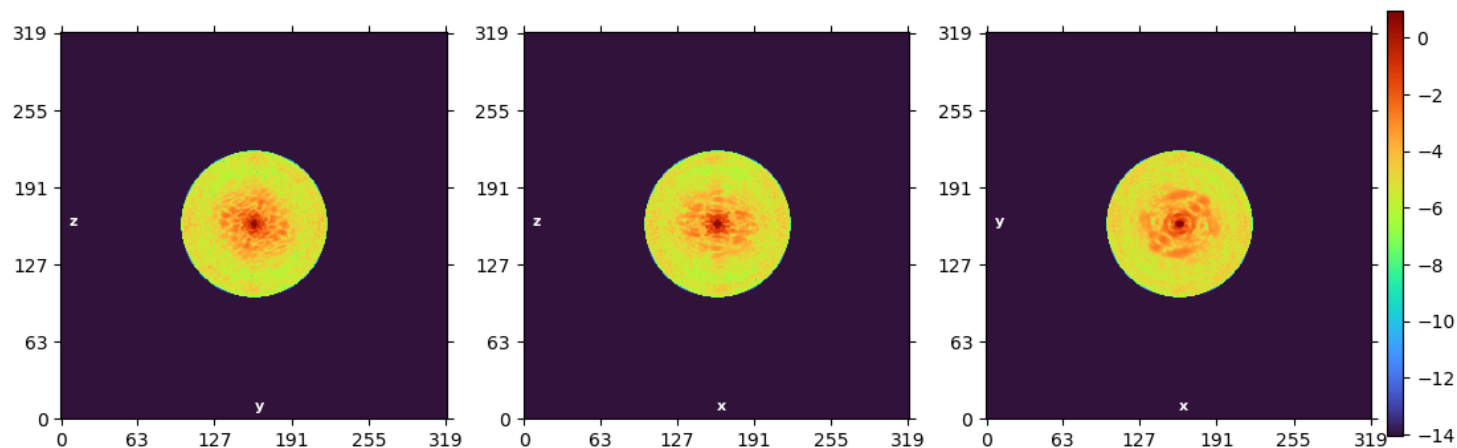
> [2026-06-05 17:45:15.56 GMT-4] [CPU: 10.28 GB] [Avail: 306.47 GB] Estimated Bfactor: -137.2

> [2026-06-05 17:45:15.56 GMT-4] [CPU: 10.28 GB] [Avail: 306.45 GB] Plotting..

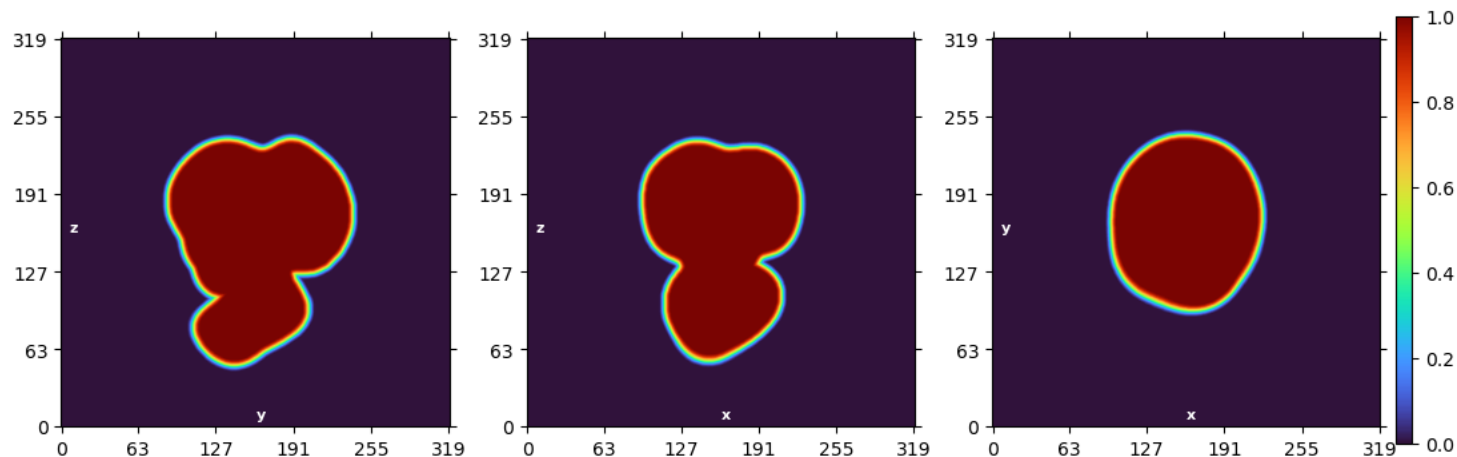
> [2026-06-05 17:45:22.49 GMT-4] Real Space Slices Iteration 001



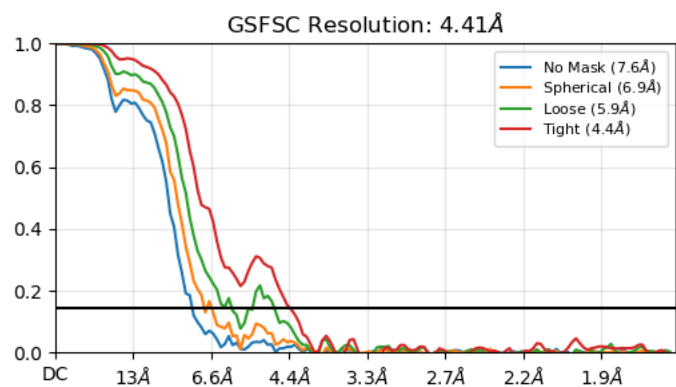
> [2026-06-05 17:45:23.98 GMT-4] Fourier Space Slices Iteration 001



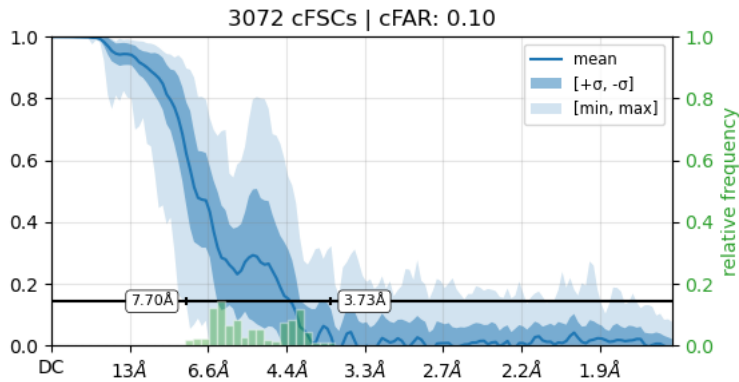
> [2026-06-05 17:45:26.21 GMT-4] Real Space Mask Slices Iteration 001



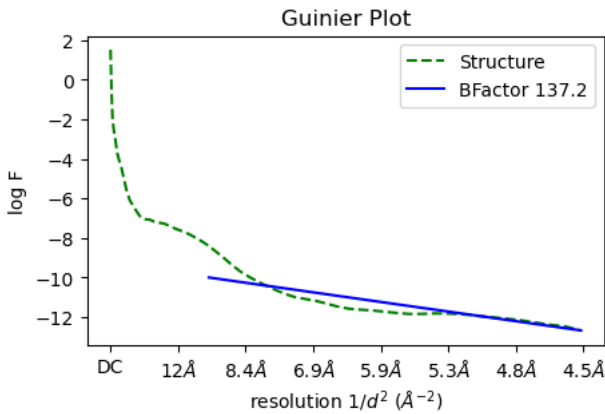
> [2026-06-05 17:45:26.82 GMT-4] FSC Iteration 001



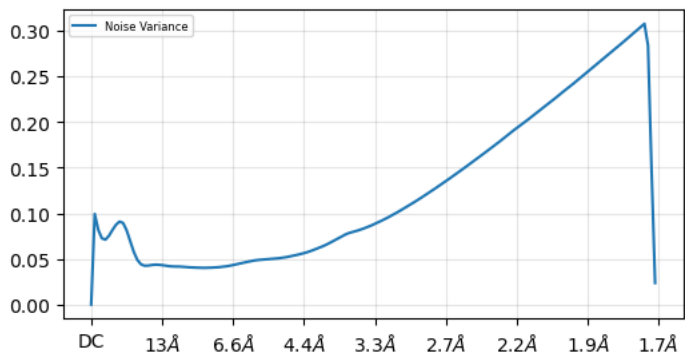
> [2026-06-05 17:45:27.70 GMT-4] cFSCs (Half-angle: 20°) Iteration 001, with tight mask



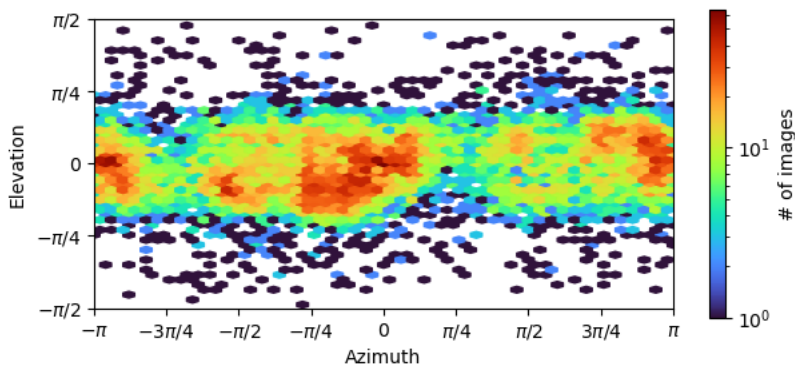
> [2026-06-05 17:45:34.03 GMT-4] Guinier Plot Iteration 001



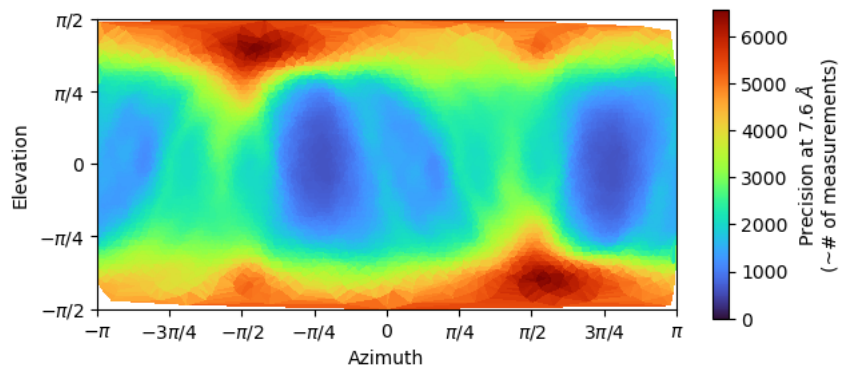
> [2026-06-05 17:45:34.47 GMT-4] Noise Model Iteration 001



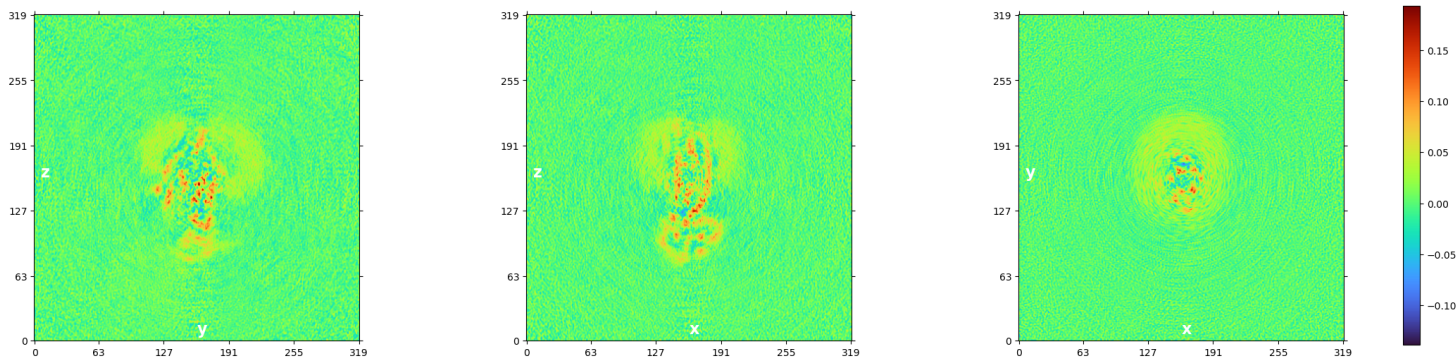
> [2026-06-05 17:45:35.47 GMT-4] Viewing Direction Distribution Iteration 001



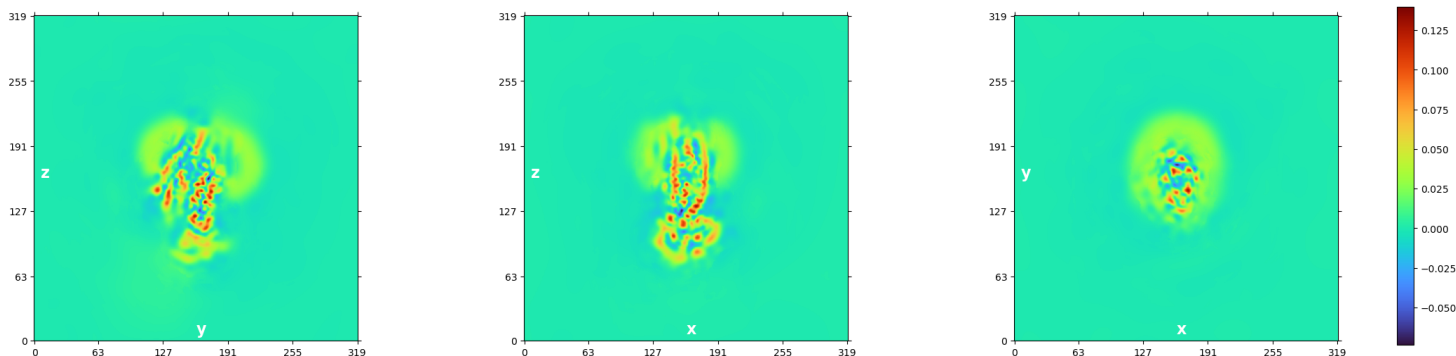
> [2026-06-05 17:45:37.78 GMT-4] Posterior Precision Directional Distribution Iteration 001



```
> [2026-06-05 17:45:37.78 GMT-4] [CPU: 10.31 GB] [Avail: 306.95 GB] Done in 22.218s.
> [2026-06-05 17:45:37.79 GMT-4] [CPU: 10.31 GB] [Avail: 306.94 GB] Outputting files..
> [2026-06-05 17:45:53.84 GMT-4] [CPU: 10.22 GB] [Avail: 300.38 GB] Done in 16.045s.
> [2026-06-05 17:45:53.84 GMT-4] [CPU: 10.22 GB] [Avail: 300.37 GB] Done iteration 1 in 178.334s. Total time so far 364.030s
> [2026-06-05 17:45:54.60 GMT-4] [CPU: 10.22 GB] [Avail: 300.72 GB] ----- Start Iteration 2
> [2026-06-05 17:45:54.60 GMT-4] [CPU: 10.22 GB] [Avail: 300.72 GB] Using Max Alignment Radius 60.277 (4.406Å)
> [2026-06-05 17:45:54.61 GMT-4] [CPU: 10.22 GB] [Avail: 300.72 GB] Auto batchsize: 6653 in each split
> [2026-06-05 17:45:54.61 GMT-4] [CPU: 10.22 GB] [Avail: 300.72 GB] Using dynamic mask.
> [2026-06-05 17:46:25.24 GMT-4] [CPU: 10.22 GB] [Avail: 303.49 GB] -- THR 0 BATCH 500 NUM 1779 TOTAL 4.1430680 ELAPSED 33.196386
--
> [2026-06-05 17:47:00.81 GMT-4] [CPU: 11.07 GB] [Avail: 294.80 GB] Processed 13211.000 images in 35.567s.
> [2026-06-05 17:47:03.92 GMT-4] [CPU: 11.07 GB] [Avail: 294.71 GB] Computing FSCs...
> [2026-06-05 17:47:03.93 GMT-4] [CPU: 11.07 GB] [Avail: 294.71 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 17:47:03.93 GMT-4] [CPU: 11.07 GB] [Avail: 294.71 GB] Computing FFTs on GPU.
> [2026-06-05 17:47:05.83 GMT-4] [CPU: 11.07 GB] [Avail: 294.81 GB] Done in 1.906s
> [2026-06-05 17:47:05.84 GMT-4] [CPU: 11.07 GB] [Avail: 294.81 GB] Computing cFSCs...
> [2026-06-05 17:47:07.80 GMT-4] [CPU: 11.07 GB] [Avail: 296.26 GB] Done in 1.961s
> [2026-06-05 17:47:07.81 GMT-4] [CPU: 11.07 GB] [Avail: 296.28 GB] Using Filter Radius 71.698 (3.704Å) | Previous: 60.277
(4.406Å)
> [2026-06-05 17:47:13.93 GMT-4] [CPU: 10.31 GB] [Avail: 297.14 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 17:47:13.93 GMT-4] [CPU: 10.31 GB] [Avail: 297.13 GB] Running local cross validation for A ...
> [2026-06-05 17:47:36.54 GMT-4] [CPU: 10.57 GB] [Avail: 293.80 GB] Local cross validation A done in 22.607s
> [2026-06-05 17:47:38.14 GMT-4] FSC Filtered Side A
```



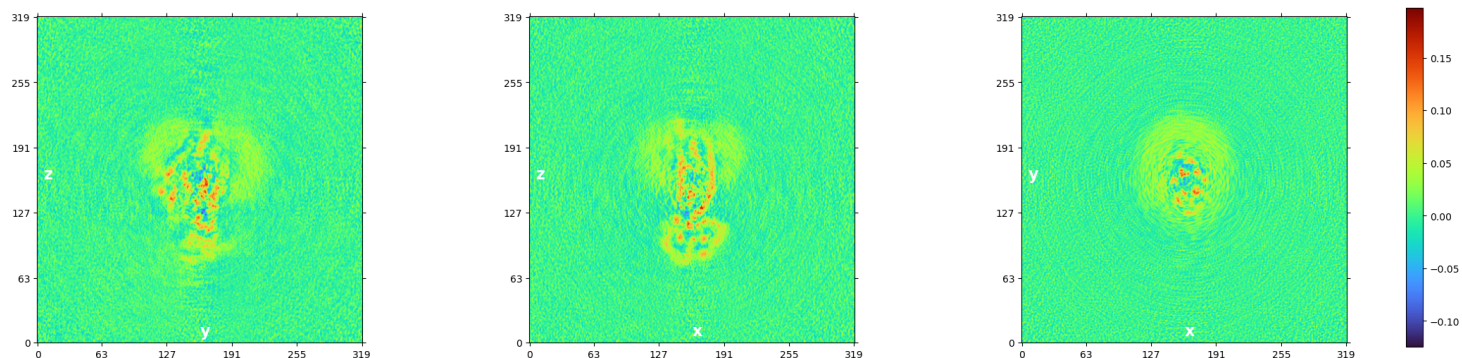
> [2026-06-05 17:47:39.50 GMT-4] CV Filtered Side A



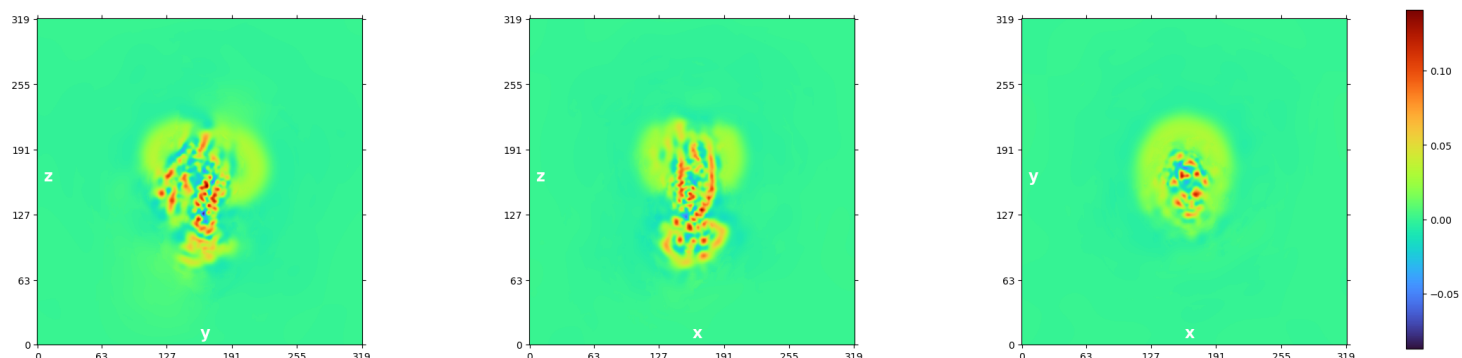
```
> [2026-06-05 17:47:39.50 GMT-4] [CPU: 10.57 GB] [Avail: 293.76 GB] Running local cross validation for B ...
```

> [2026-06-05 17:48:01.38 GMT-4] [CPU: 10.82 GB] [Avail: 298.43 GB] Local cross validation B done in 21.870s

> [2026-06-05 17:48:02.72 GMT-4] FSC Filtered Side B



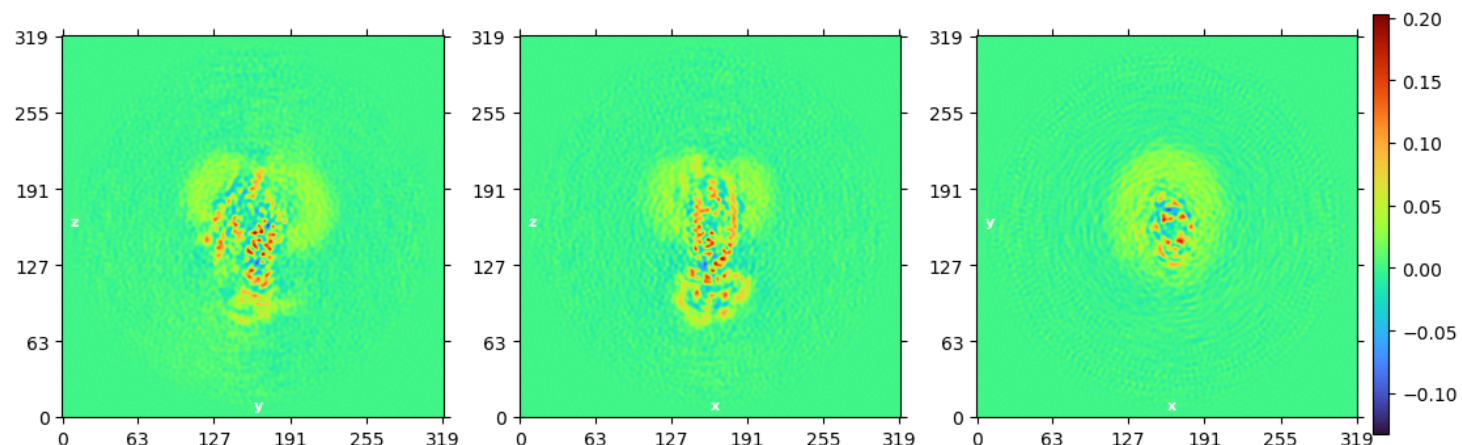
> [2026-06-05 17:48:03.82 GMT-4] CV Filtered Side B



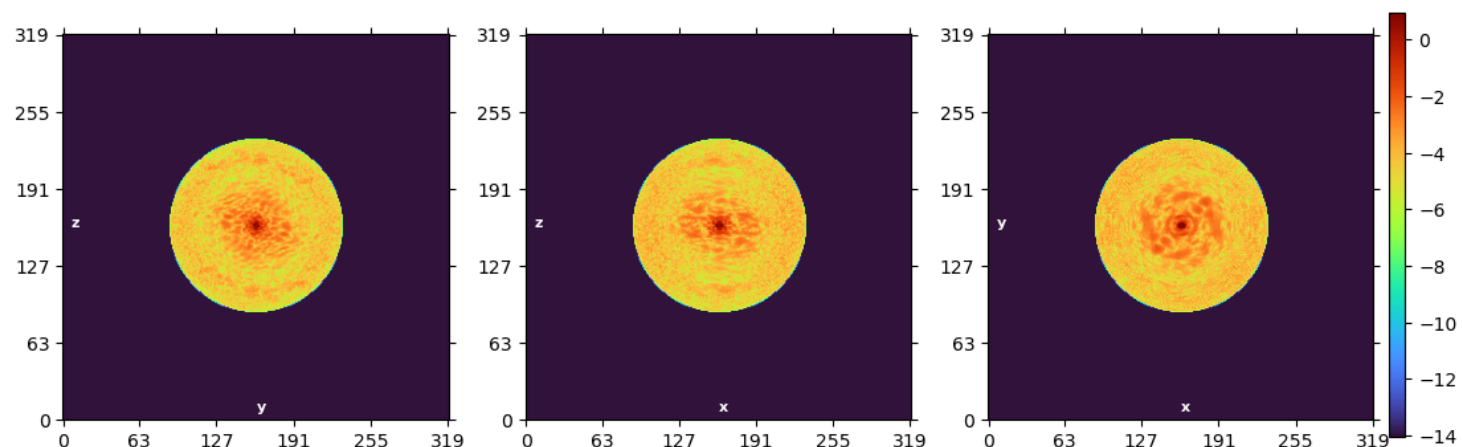
> [2026-06-05 17:48:14.01 GMT-4] [CPU: 10.82 GB] [Avail: 294.59 GB] Estimated Bfactor: -87.6

> [2026-06-05 17:48:14.02 GMT-4] [CPU: 10.82 GB] [Avail: 294.59 GB] Plotting..

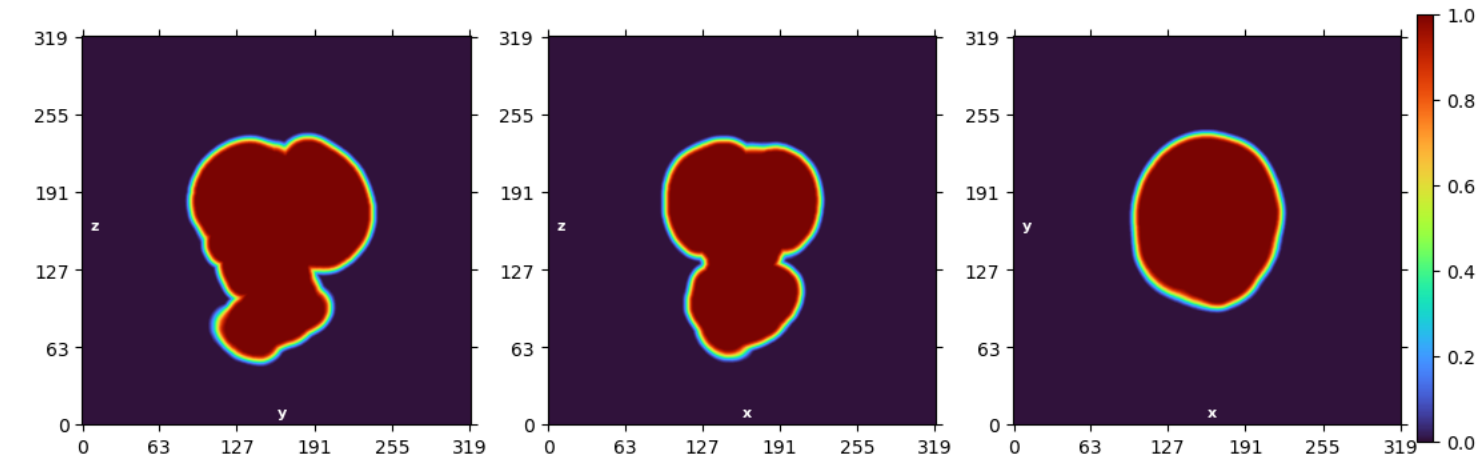
> [2026-06-05 17:48:20.09 GMT-4] Real Space Slices Iteration 002



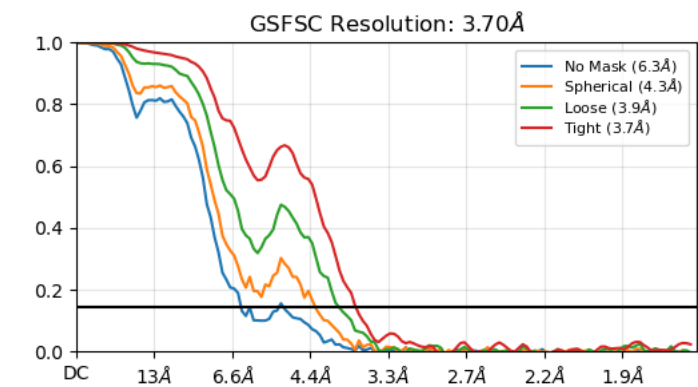
> [2026-06-05 17:48:21.36 GMT-4] Fourier Space Slices Iteration 002



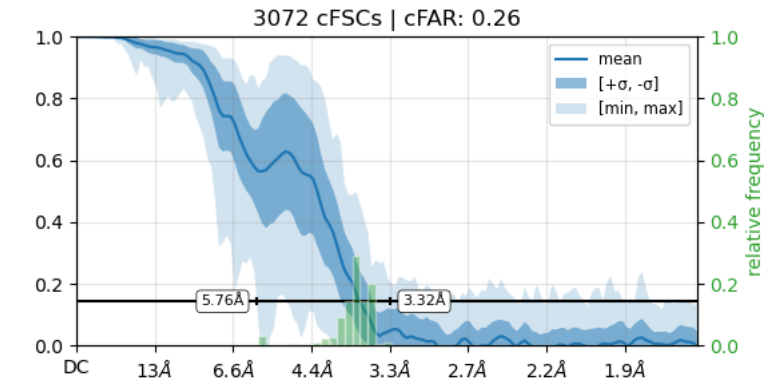
> [2026-06-05 17:48:23.29 GMT-4] Real Space Mask Slices Iteration 002



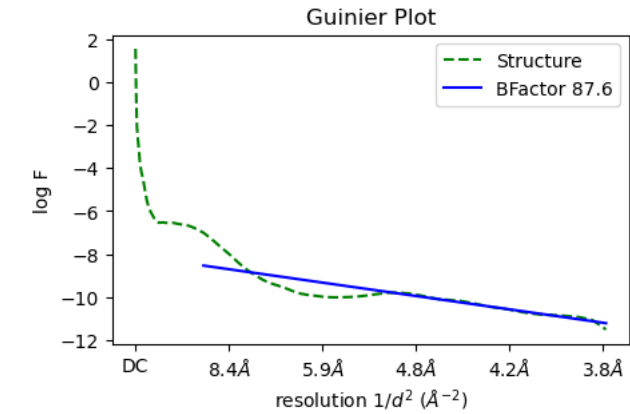
> [2026-06-05 17:48:23.81 GMT-4] FSC Iteration 002



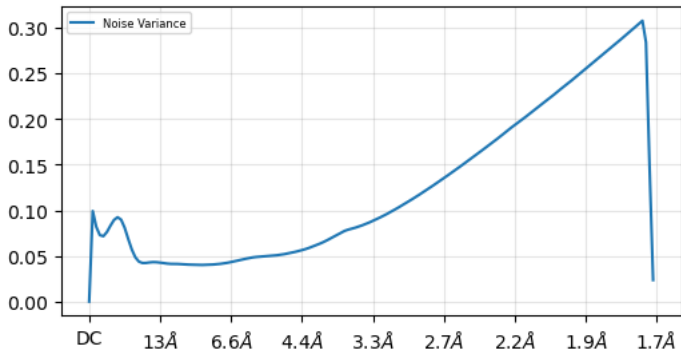
> [2026-06-05 17:48:24.60 GMT-4] cFSCs (Half-angle: 20°) Iteration 002, with tight mask



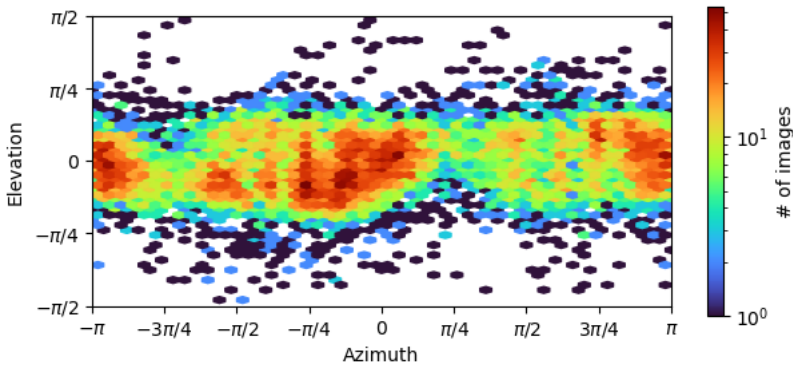
> [2026-06-05 17:48:30.66 GMT-4] Guinier Plot Iteration 002



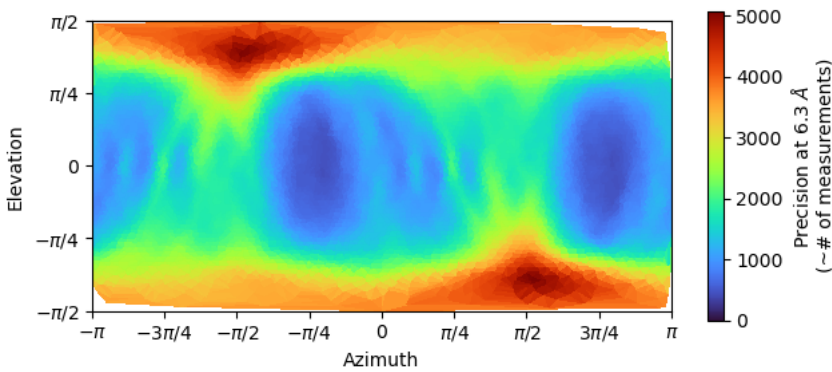
> [2026-06-05 17:48:31.04 GMT-4] Noise Model Iteration 002



> [2026-06-05 17:48:31.86 GMT-4] Viewing Direction Distribution Iteration 002

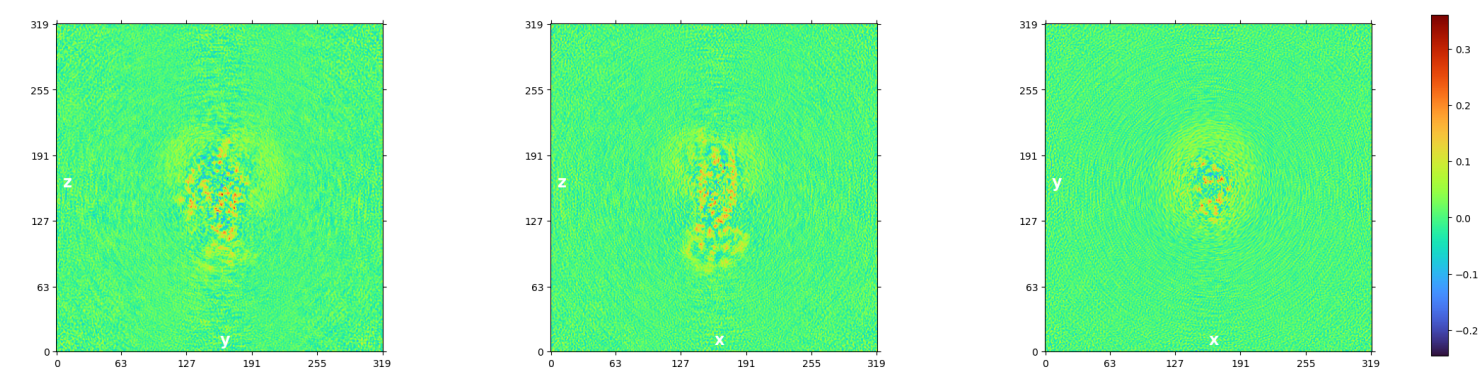


> [2026-06-05 17:48:33.49 GMT-4] Posterior Precision Directional Distribution Iteration 002

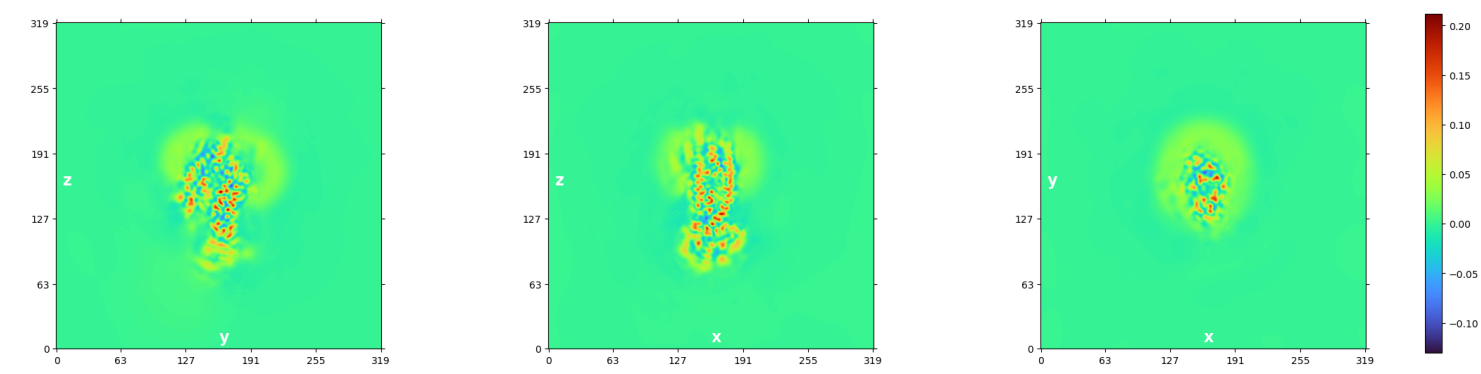


```
> [2026-06-05 17:48:33.50 GMT-4] [CPU: 10.82 GB] [Avail: 293.73 GB] Done in 19.476s.
> [2026-06-05 17:48:33.50 GMT-4] [CPU: 10.82 GB] [Avail: 293.74 GB] Outputting files..
> [2026-06-05 17:48:49.49 GMT-4] [CPU: 10.72 GB] [Avail: 293.40 GB] Done in 15.977s.
> [2026-06-05 17:48:49.49 GMT-4] [CPU: 10.72 GB] [Avail: 293.39 GB] Done iteration 2 in 174.884s. Total time so far 539.679s
> [2026-06-05 17:48:50.29 GMT-4] [CPU: 10.72 GB] [Avail: 293.19 GB] ----- Start Iteration 3
> [2026-06-05 17:48:50.30 GMT-4] [CPU: 10.72 GB] [Avail: 293.19 GB] Using Max Alignment Radius 71.698 (3.704A)
> [2026-06-05 17:48:50.30 GMT-4] [CPU: 10.72 GB] [Avail: 293.18 GB] Auto batchsize: 6653 in each split
> [2026-06-05 17:48:50.30 GMT-4] [CPU: 10.72 GB] [Avail: 293.18 GB] Using dynamic mask.
> [2026-06-05 17:49:18.58 GMT-4] [CPU: 10.72 GB] [Avail: 298.25 GB] -- THR 1 BATCH 500 NUM 1779 TOTAL 4.0913751 ELAPSED 32.633754
--
> [2026-06-05 17:49:53.22 GMT-4] [CPU: 11.48 GB] [Avail: 292.56 GB] Processed 13211.000 images in 34.640s.
> [2026-06-05 17:49:56.39 GMT-4] [CPU: 11.51 GB] [Avail: 292.52 GB] Computing FSCs...
> [2026-06-05 17:49:56.39 GMT-4] [CPU: 11.51 GB] [Avail: 292.52 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 17:49:56.40 GMT-4] [CPU: 11.51 GB] [Avail: 292.51 GB] Computing FFTs on GPU.
> [2026-06-05 17:49:58.34 GMT-4] [CPU: 11.51 GB] [Avail: 292.12 GB] Done in 1.949s
> [2026-06-05 17:49:58.35 GMT-4] [CPU: 11.51 GB] [Avail: 292.11 GB] Computing cFSCs...
> [2026-06-05 17:50:00.38 GMT-4] [CPU: 11.51 GB] [Avail: 291.93 GB] Done in 2.032s
> [2026-06-05 17:50:00.39 GMT-4] [CPU: 11.51 GB] [Avail: 291.97 GB] Using Filter Radius 77.677 (3.419A) | Previous: 71.698
(3.704A)
> [2026-06-05 17:50:06.55 GMT-4] [CPU: 10.76 GB] [Avail: 297.25 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 17:50:06.55 GMT-4] [CPU: 10.76 GB] [Avail: 297.24 GB] Running local cross validation for A ...
> [2026-06-05 17:50:28.83 GMT-4] [CPU: 10.98 GB] [Avail: 294.08 GB] Local cross validation A done in 22.271s
```

> [2026-06-05 17:50:30.22 GMT-4] FSC Filtered Side A



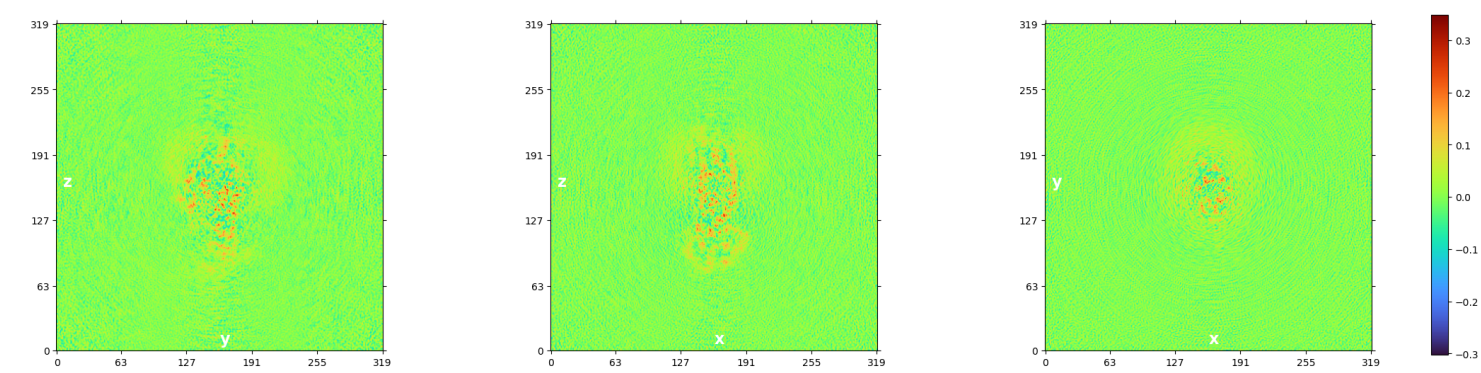
> [2026-06-05 17:50:31.28 GMT-4] CV Filtered Side A



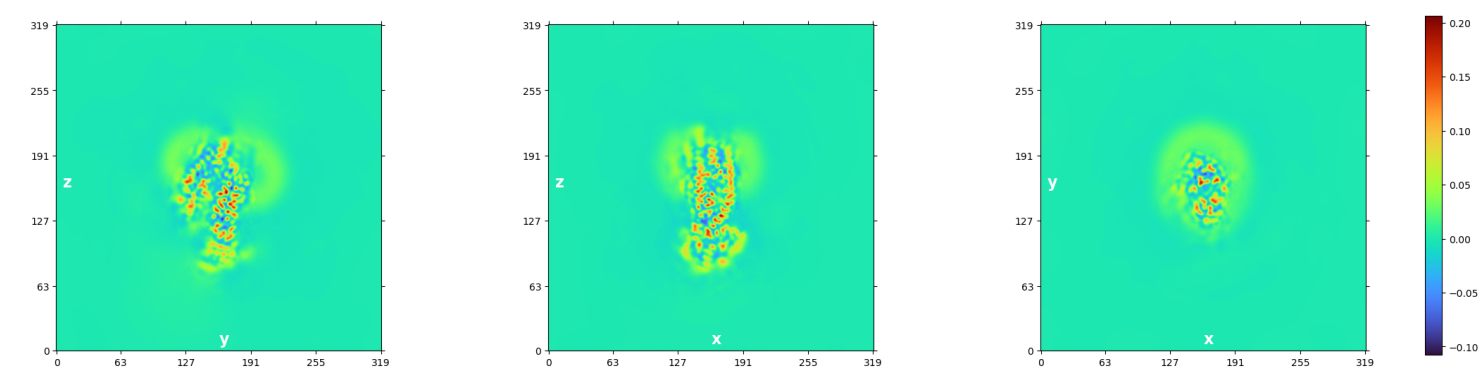
> [2026-06-05 17:50:31.28 GMT-4] [CPU: 10.98 GB] [Avail: 293.85 GB] Running local cross validation for B ...

> [2026-06-05 17:50:53.02 GMT-4] [CPU: 11.23 GB] [Avail: 300.54 GB] Local cross validation B done in 21.734s

> [2026-06-05 17:50:54.41 GMT-4] FSC Filtered Side B



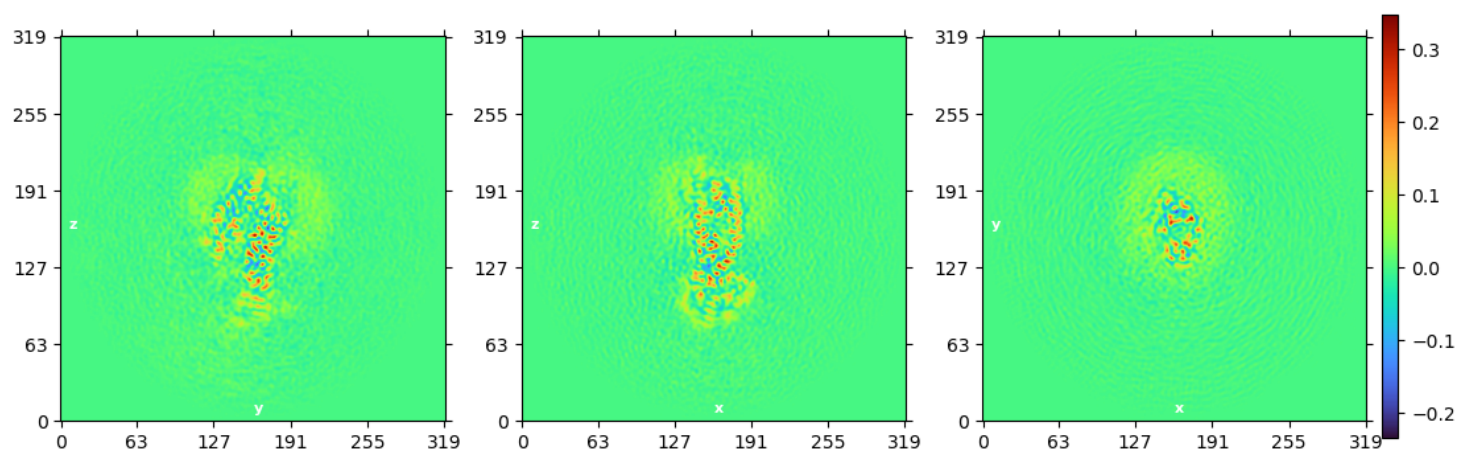
> [2026-06-05 17:50:55.44 GMT-4] CV Filtered Side B



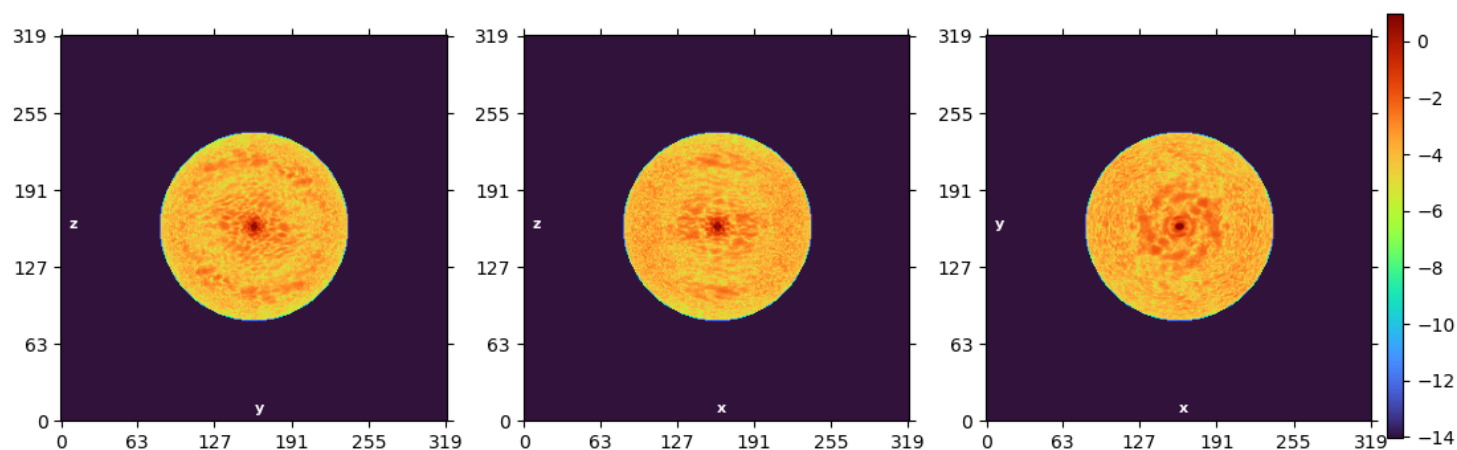
> [2026-06-05 17:51:05.50 GMT-4] [CPU: 11.29 GB] [Avail: 296.07 GB] Estimated Bfactor: -70.3

> [2026-06-05 17:51:05.51 GMT-4] [CPU: 11.29 GB] [Avail: 296.06 GB] Plotting..

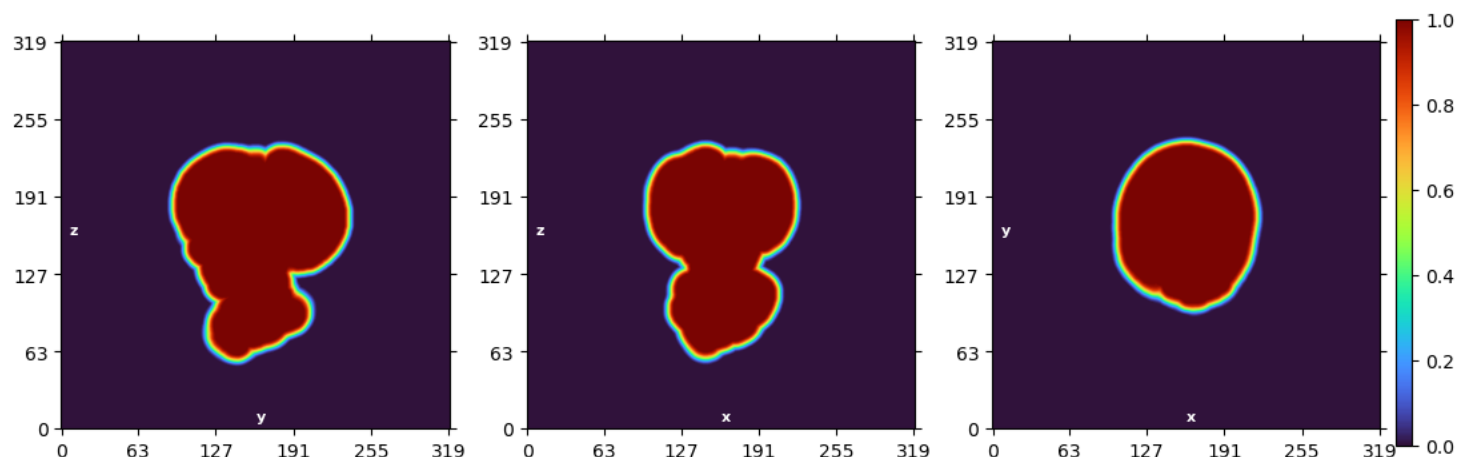
> [2026-06-05 17:51:12.79 GMT-4] Real Space Slices Iteration 003



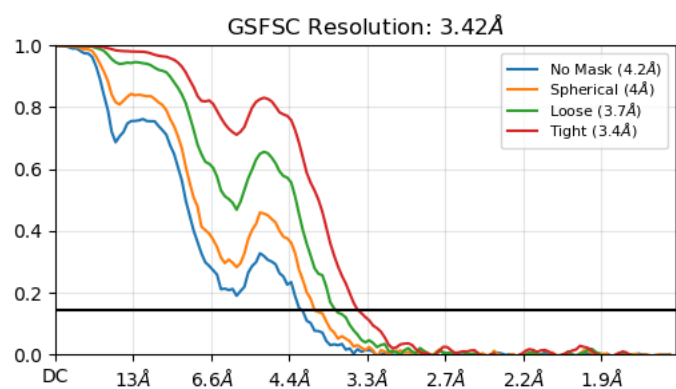
> [2026-06-05 17:51:14.11 GMT-4] Fourier Space Slices Iteration 003



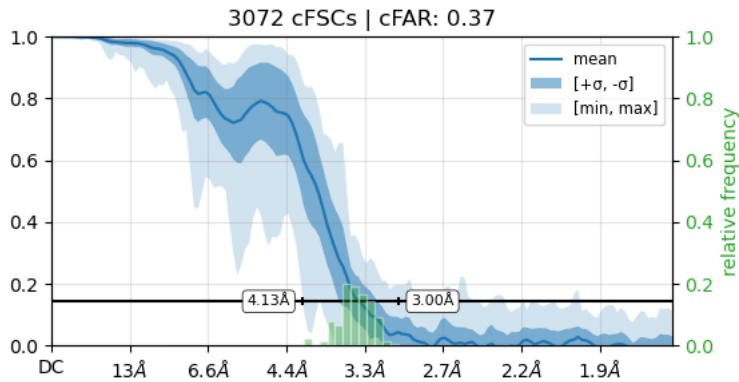
> [2026-06-05 17:51:16.19 GMT-4] Real Space Mask Slices Iteration 003



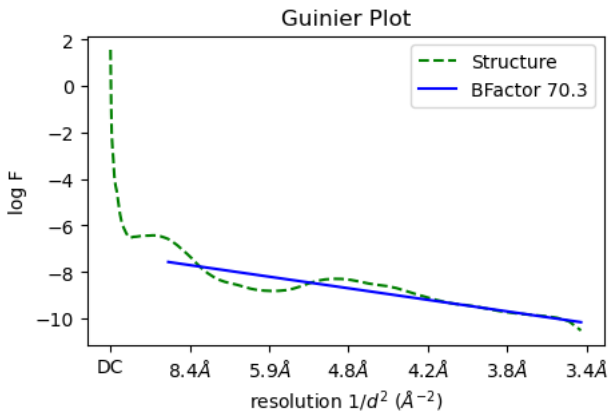
> [2026-06-05 17:51:16.73 GMT-4] FSC Iteration 003



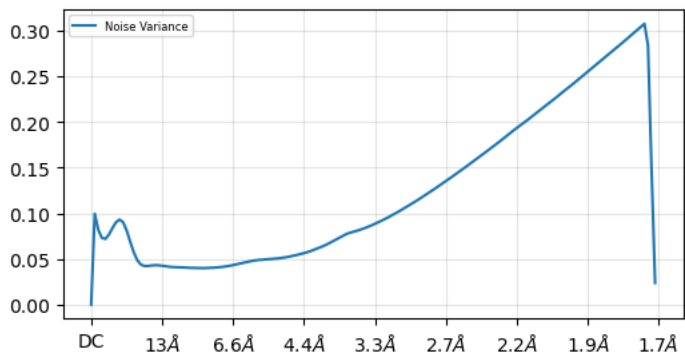
> [2026-06-05 17:51:17.51 GMT-4] cFSCs (Half-angle: 20°) Iteration 003, with tight mask



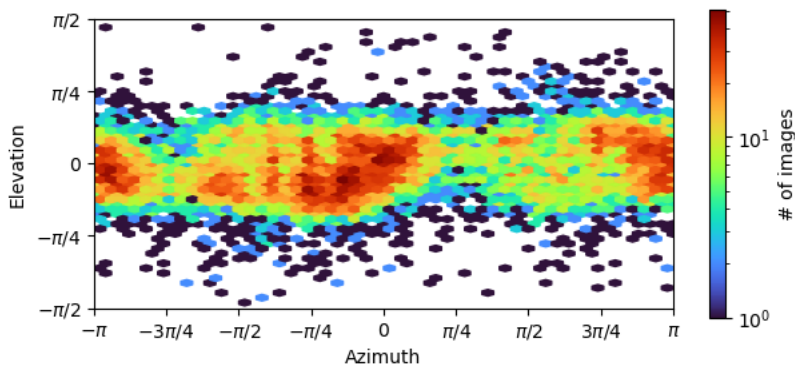
> [2026-06-05 17:51:23.99 GMT-4] Guinier Plot Iteration 003



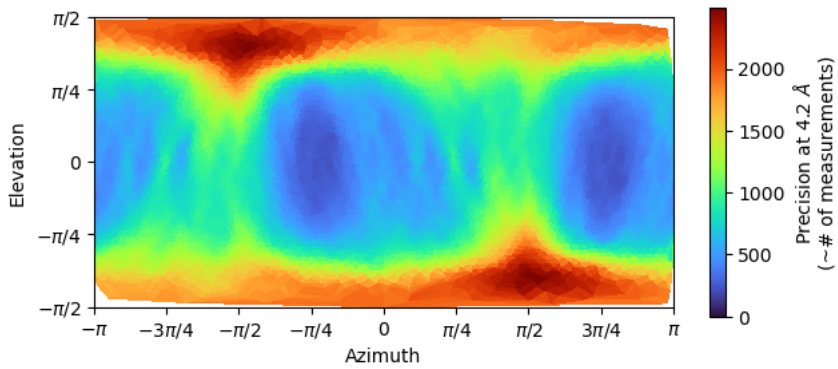
> [2026-06-05 17:51:24.34 GMT-4] Noise Model Iteration 003



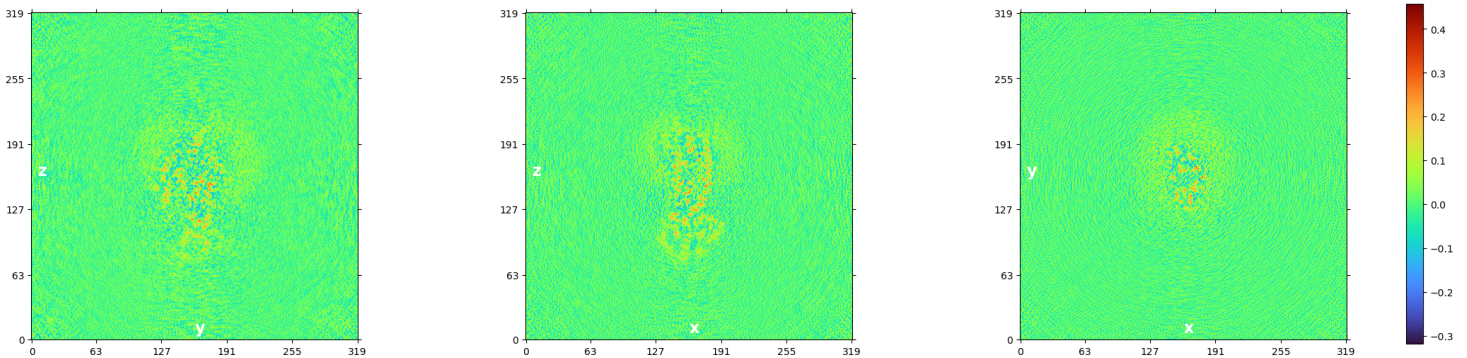
> [2026-06-05 17:51:25.31 GMT-4] Viewing Direction Distribution Iteration 003



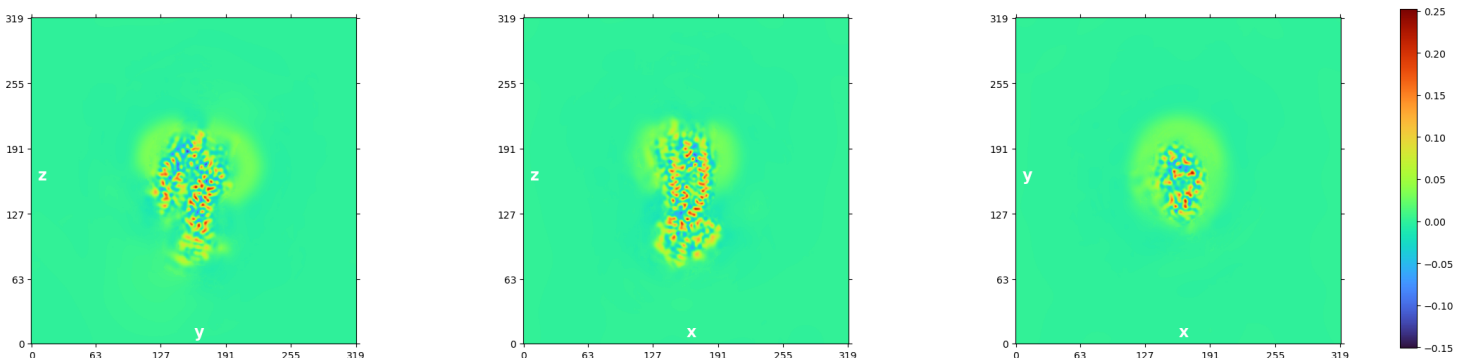
> [2026-06-05 17:51:26.93 GMT-4] Posterior Precision Directional Distribution Iteration 003



```
> [2026-06-05 17:51:26.93 GMT-4] [CPU: 11.29 GB] [Avail: 297.30 GB] Done in 21.418s.
> [2026-06-05 17:51:26.94 GMT-4] [CPU: 11.29 GB] [Avail: 297.30 GB] Outputting files..
> [2026-06-05 17:51:42.80 GMT-4] [CPU: 11.17 GB] [Avail: 296.93 GB] Done in 15.862s.
> [2026-06-05 17:51:42.80 GMT-4] [CPU: 11.17 GB] [Avail: 296.93 GB] Done iteration 3 in 172.509s. Total time so far 712.994s
> [2026-06-05 17:51:43.57 GMT-4] [CPU: 11.17 GB] [Avail: 297.65 GB] ----- Start Iteration 4
> [2026-06-05 17:51:43.57 GMT-4] [CPU: 11.17 GB] [Avail: 297.66 GB] Using Max Alignment Radius 77.677 (3.419A)
> [2026-06-05 17:51:43.57 GMT-4] [CPU: 11.17 GB] [Avail: 297.66 GB] Auto batchsize: 6653 in each split
> [2026-06-05 17:51:43.58 GMT-4] [CPU: 11.17 GB] [Avail: 297.67 GB] Using dynamic mask.
> [2026-06-05 17:52:12.04 GMT-4] [CPU: 11.17 GB] [Avail: 304.15 GB] -- THR 0 BATCH 500 NUM 1779 TOTAL 3.4201383 ELAPSED 33.108624
--
> [2026-06-05 17:52:47.05 GMT-4] [CPU: 11.92 GB] [Avail: 296.87 GB] Processed 13211.000 images in 35.016s.
> [2026-06-05 17:52:50.49 GMT-4] [CPU: 11.95 GB] [Avail: 298.06 GB] Computing FSCs...
> [2026-06-05 17:52:50.50 GMT-4] [CPU: 11.95 GB] [Avail: 298.06 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 17:52:50.50 GMT-4] [CPU: 11.95 GB] [Avail: 298.06 GB] Computing FFTs on GPU.
> [2026-06-05 17:52:52.52 GMT-4] [CPU: 11.95 GB] [Avail: 298.50 GB] Done in 2.024s
> [2026-06-05 17:52:52.52 GMT-4] [CPU: 11.95 GB] [Avail: 298.49 GB] Computing cFSCs...
> [2026-06-05 17:52:54.72 GMT-4] [CPU: 11.95 GB] [Avail: 296.17 GB] Done in 2.195s
> [2026-06-05 17:52:54.73 GMT-4] [CPU: 11.95 GB] [Avail: 296.16 GB] Using Filter Radius 79.923 (3.323A) | Previous: 77.677
(3.419A)
> [2026-06-05 17:53:01.32 GMT-4] [CPU: 11.20 GB] [Avail: 298.79 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 17:53:01.33 GMT-4] [CPU: 11.20 GB] [Avail: 298.79 GB] Running local cross validation for A ...
> [2026-06-05 17:53:25.81 GMT-4] [CPU: 11.42 GB] [Avail: 290.09 GB] Local cross validation A done in 24.479s
> [2026-06-05 17:53:27.31 GMT-4] FSC Filtered Side A
```



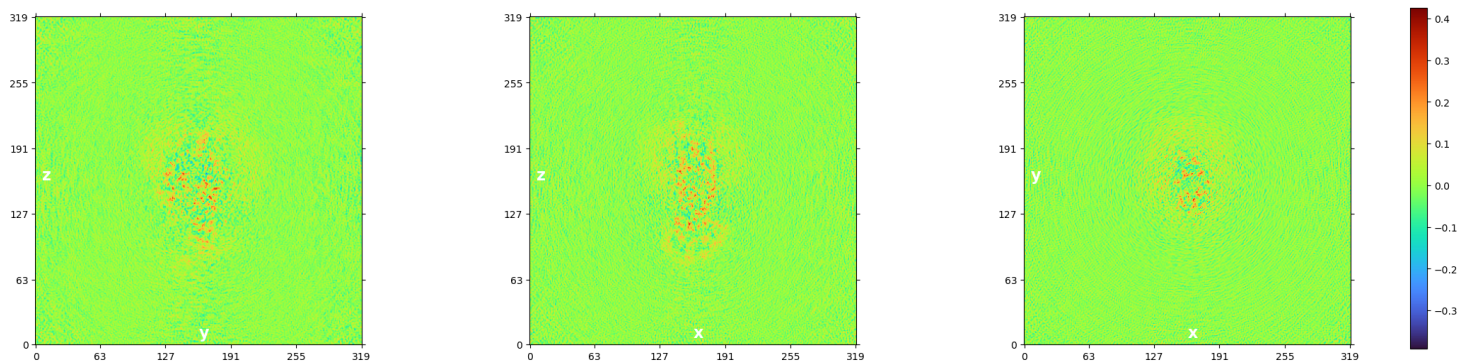
> [2026-06-05 17:53:28.45 GMT-4] CV Filtered Side A



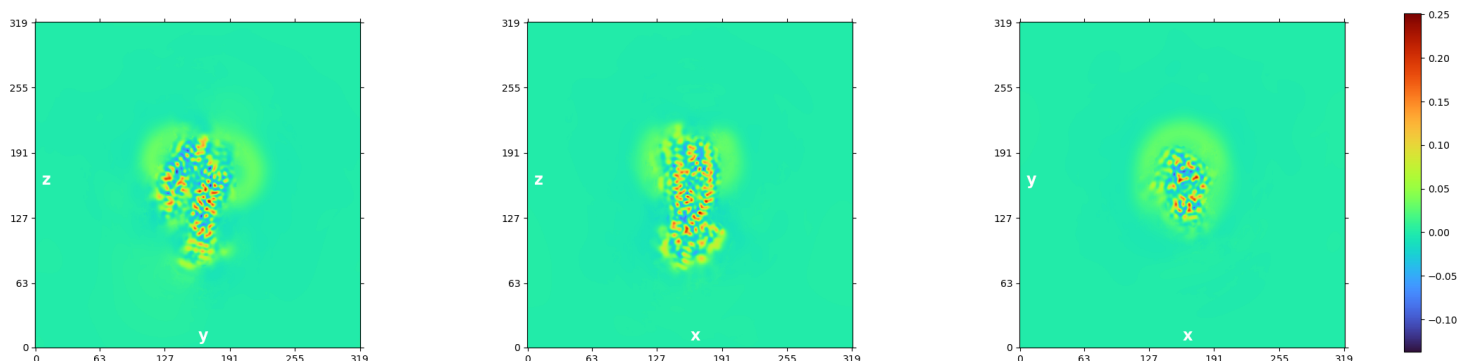
```
> [2026-06-05 17:53:28.45 GMT-4] [CPU: 11.42 GB] [Avail: 294.74 GB] Running local cross validation for B ...
```

> [2026-06-05 17:53:53.30 GMT-4] [CPU: 11.67 GB] [Avail: 297.92 GB] Local cross validation B done in 24.843s

> [2026-06-05 17:53:54.80 GMT-4] FSC Filtered Side B



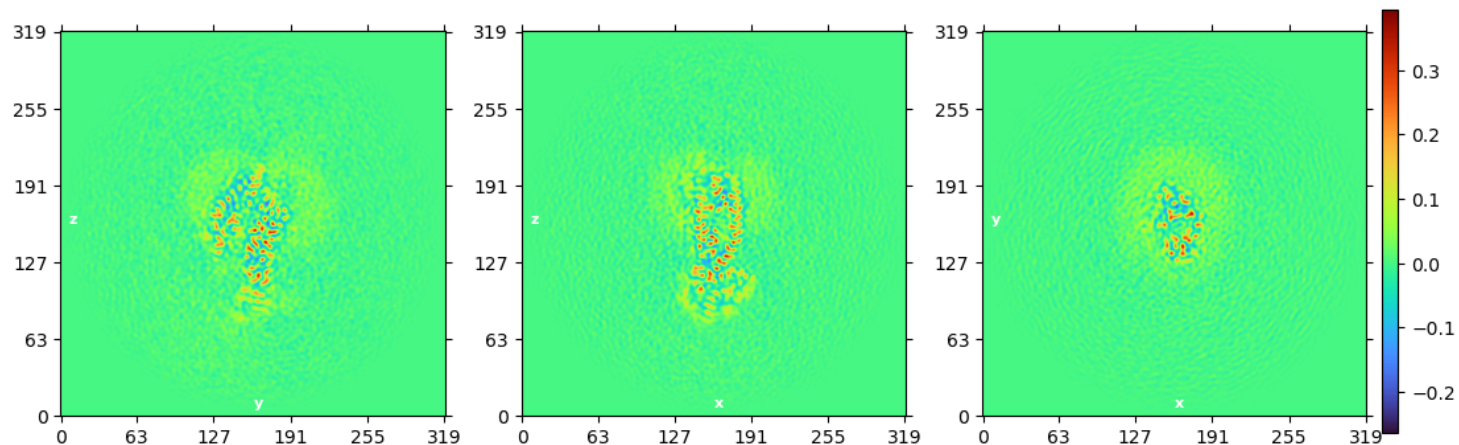
> [2026-06-05 17:53:56.03 GMT-4] CV Filtered Side B



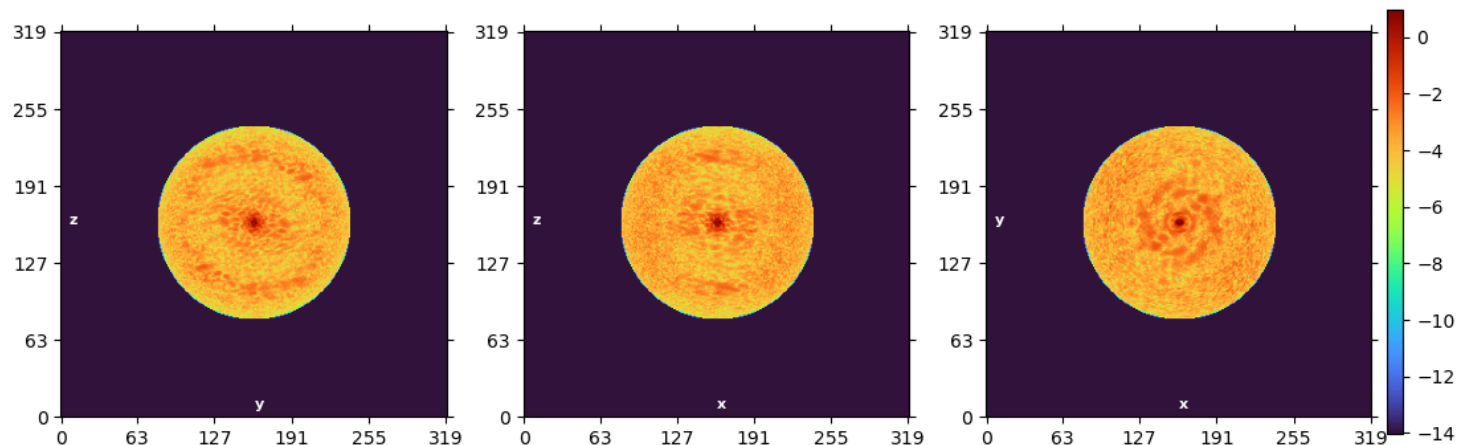
> [2026-06-05 17:54:07.39 GMT-4] [CPU: 11.67 GB] [Avail: 291.02 GB] Estimated Bfactor: -59.3

> [2026-06-05 17:54:07.40 GMT-4] [CPU: 11.67 GB] [Avail: 291.01 GB] Plotting..

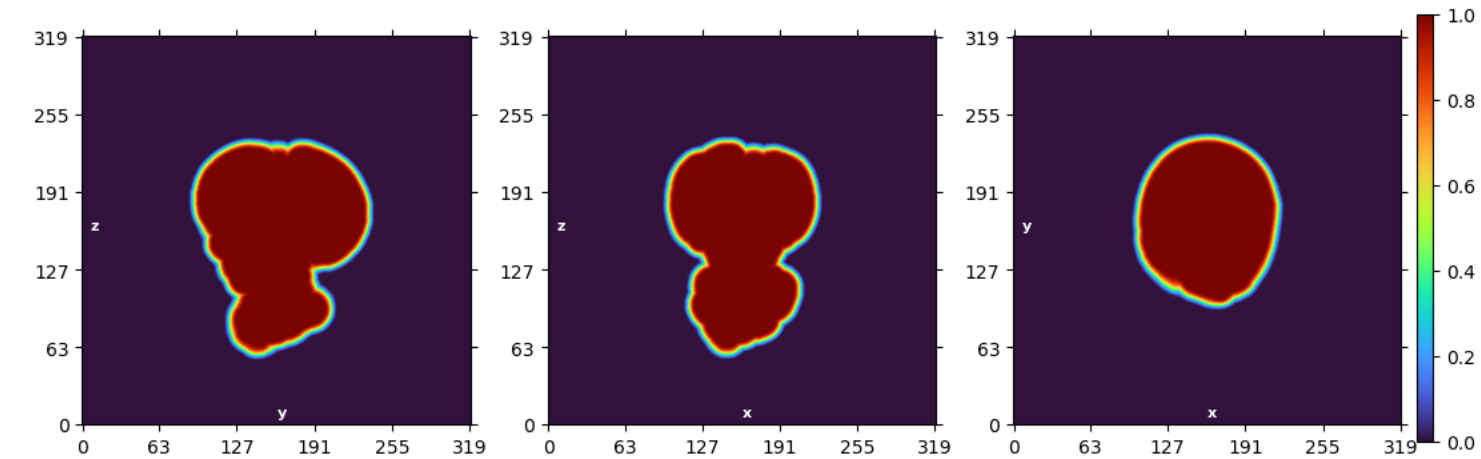
> [2026-06-05 17:54:14.11 GMT-4] Real Space Slices Iteration 004



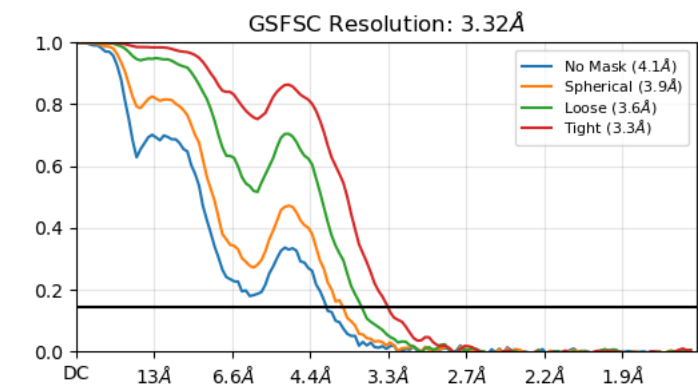
> [2026-06-05 17:54:15.45 GMT-4] Fourier Space Slices Iteration 004



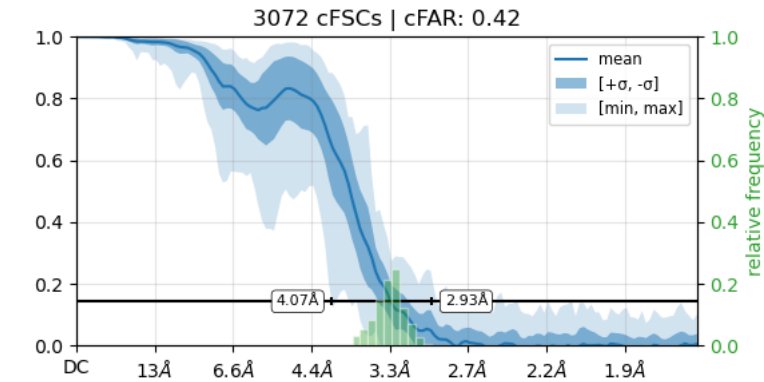
> [2026-06-05 17:54:17.54 GMT-4] Real Space Mask Slices Iteration 004



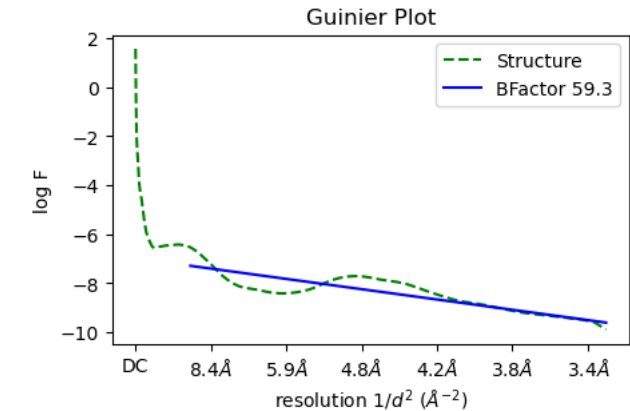
> [2026-06-05 17:54:18.05 GMT-4] FSC Iteration 004



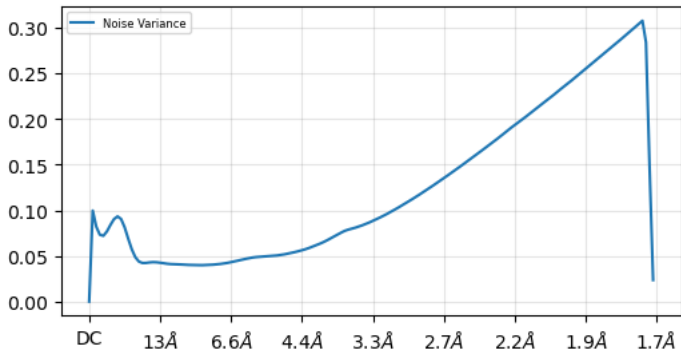
> [2026-06-05 17:54:18.87 GMT-4] cFSCs (Half-angle: 20°) Iteration 004, with tight mask



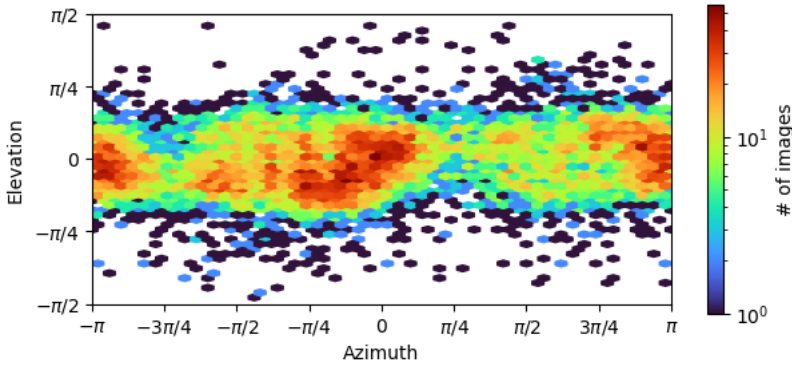
> [2026-06-05 17:54:24.91 GMT-4] Guinier Plot Iteration 004



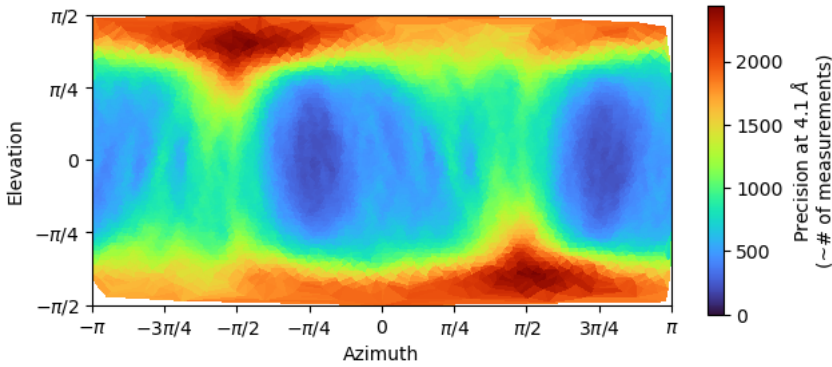
> [2026-06-05 17:54:25.29 GMT-4] Noise Model Iteration 004



> [2026-06-05 17:54:26.11 GMT-4] Viewing Direction Distribution Iteration 004



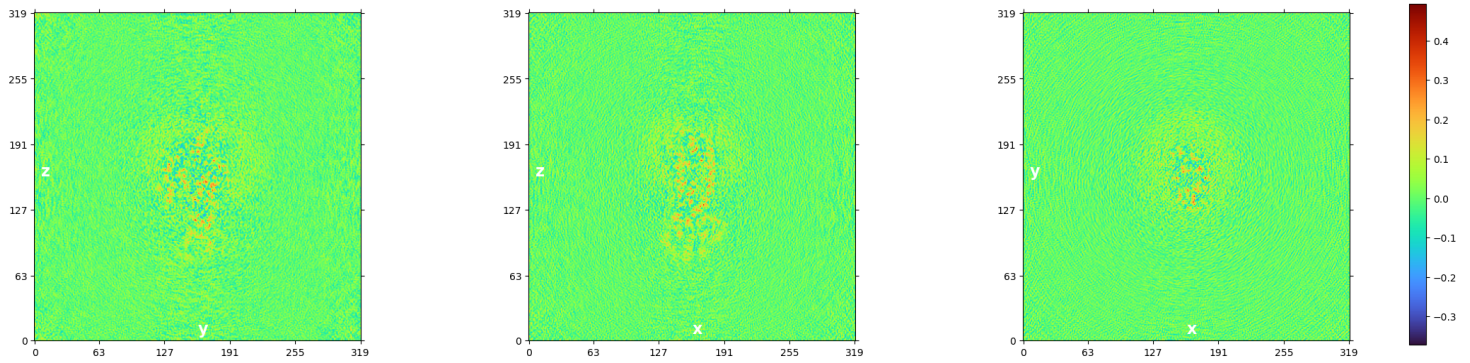
> [2026-06-05 17:54:27.71 GMT-4] Posterior Precision Directional Distribution Iteration 004



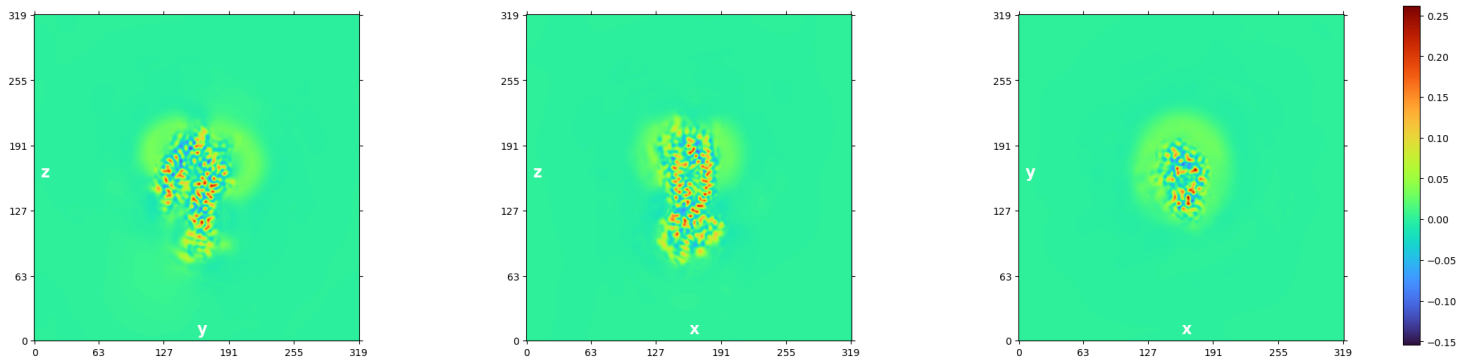
```
> [2026-06-05 17:54:27.71 GMT-4] [CPU: 11.70 GB] [Avail: 294.12 GB] Done in 20.306s.
> [2026-06-05 17:54:27.72 GMT-4] [CPU: 11.70 GB] [Avail: 294.12 GB] Outputting files..
> [2026-06-05 17:54:43.91 GMT-4] [CPU: 11.61 GB] [Avail: 291.81 GB] Done in 16.185s.
> [2026-06-05 17:54:43.91 GMT-4] [CPU: 11.61 GB] [Avail: 291.79 GB] Done iteration 4 in 180.337s. Total time so far 894.099s
> [2026-06-05 17:54:44.67 GMT-4] [CPU: 11.61 GB] [Avail: 293.18 GB] ----- Start Iteration 5
> [2026-06-05 17:54:44.67 GMT-4] [CPU: 11.61 GB] [Avail: 293.18 GB] Using Max Alignment Radius 79.923 (3.323A)
> [2026-06-05 17:54:44.67 GMT-4] [CPU: 11.61 GB] [Avail: 293.18 GB] Using Full Dataset (split 6653 in A, 6558 in B)
> [2026-06-05 17:54:44.68 GMT-4] [CPU: 11.61 GB] [Avail: 293.17 GB] Using dynamic mask.
> [2026-06-05 17:55:13.74 GMT-4] [CPU: 11.61 GB] [Avail: 294.72 GB] -- THR 1 BATCH 500 NUM 1779 TOTAL 4.1047537 ELAPSED 32.726081
--
> [2026-06-05 17:55:49.20 GMT-4] [CPU: 12.36 GB] [Avail: 289.87 GB] Processed 13211.000 images in 35.457s.
> [2026-06-05 17:55:52.33 GMT-4] [CPU: 12.39 GB] [Avail: 288.11 GB] Computing FSCs...
> [2026-06-05 17:55:52.34 GMT-4] [CPU: 12.39 GB] [Avail: 288.10 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 17:55:52.34 GMT-4] [CPU: 12.39 GB] [Avail: 288.10 GB] Computing FFTs on GPU.
> [2026-06-05 17:55:55.53 GMT-4] [CPU: 12.39 GB] [Avail: 288.14 GB] Done in 3.190s
> [2026-06-05 17:55:55.53 GMT-4] [CPU: 12.39 GB] [Avail: 288.14 GB] Optimizing FSC Mask...
> [2026-06-05 17:55:55.53 GMT-4] [CPU: 12.39 GB] [Avail: 288.14 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 17:55:55.54 GMT-4] [CPU: 12.39 GB] [Avail: 288.14 GB] Computing FFTs on GPU.
> [2026-06-05 17:55:57.55 GMT-4] [CPU: 12.64 GB] [Avail: 287.70 GB] Done in 2.021s
> [2026-06-05 17:55:57.56 GMT-4] [CPU: 12.64 GB] [Avail: 287.70 GB] Computing cFSCs...
> [2026-06-05 17:55:59.62 GMT-4] [CPU: 12.65 GB] [Avail: 288.65 GB] Done in 2.062s
> [2026-06-05 17:55:59.63 GMT-4] [CPU: 12.65 GB] [Avail: 288.66 GB] Using Filter Radius 82.919 (3.203A) | Previous: 79.923
```

(3.323A)

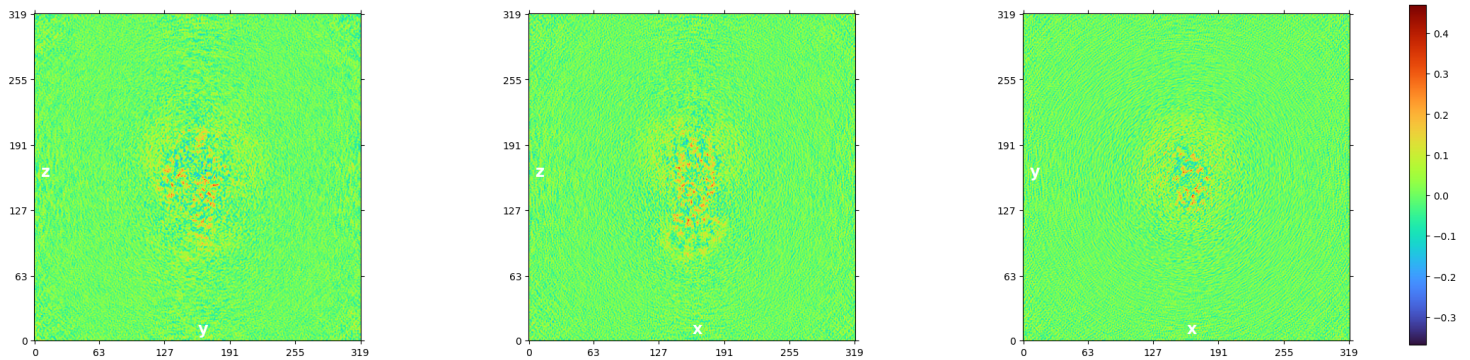
```
> [2026-06-05 17:56:06.00 GMT-4] [CPU: 11.89 GB] [Avail: 288.75 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 17:56:06.00 GMT-4] [CPU: 11.89 GB] [Avail: 288.74 GB] Running local cross validation for A ...
> [2026-06-05 17:56:30.11 GMT-4] [CPU: 12.11 GB] [Avail: 289.20 GB] Local cross validation A done in 24.101s
> [2026-06-05 17:56:31.93 GMT-4] FSC Filtered Side A
```



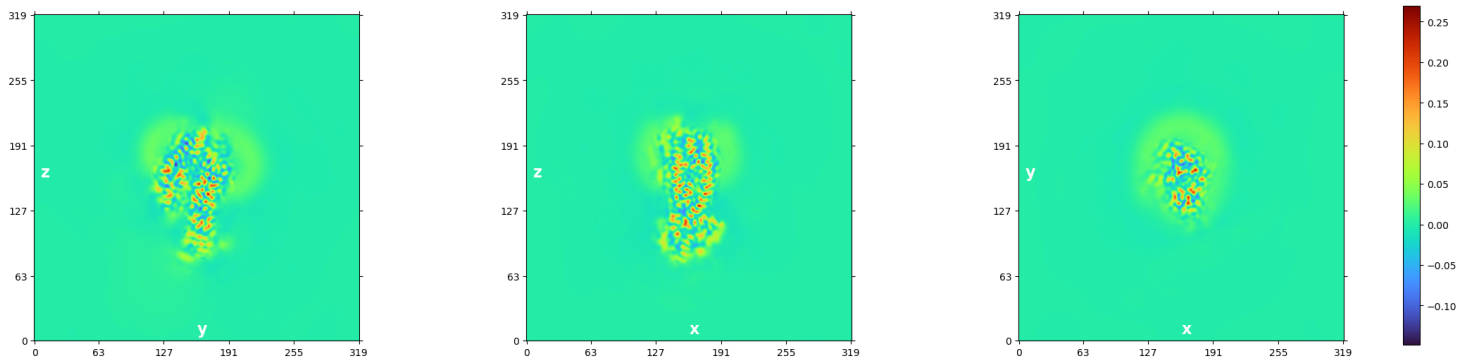
```
> [2026-06-05 17:56:33.20 GMT-4] CV Filtered Side A
```



```
> [2026-06-05 17:56:33.20 GMT-4] [CPU: 12.11 GB] [Avail: 292.85 GB] Running local cross validation for B ...
> [2026-06-05 17:56:55.72 GMT-4] [CPU: 12.36 GB] [Avail: 294.68 GB] Local cross validation B done in 22.514s
> [2026-06-05 17:56:57.21 GMT-4] FSC Filtered Side B
```

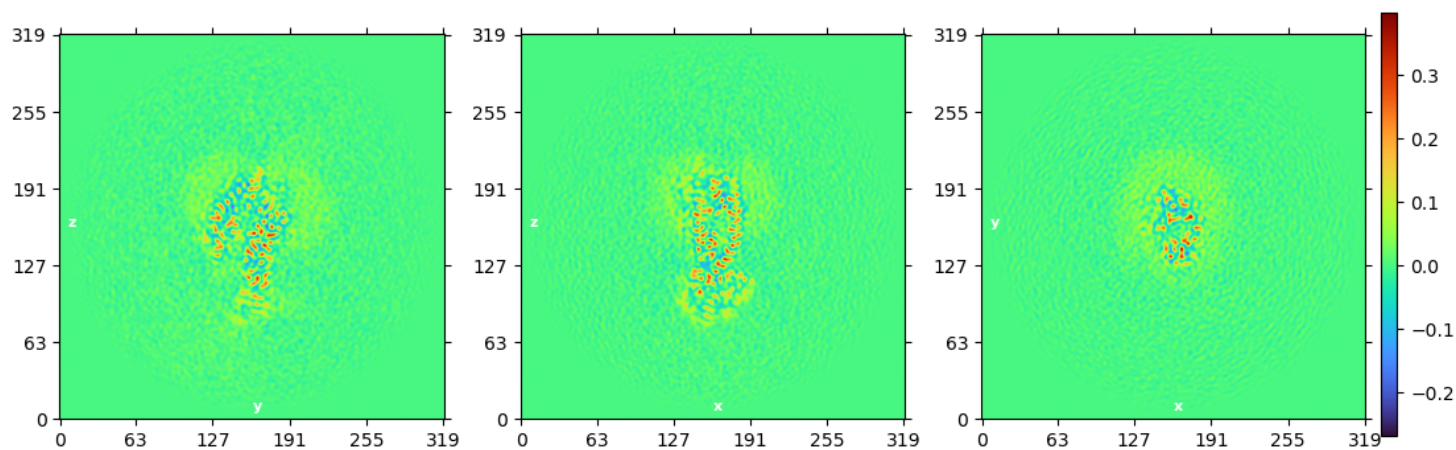


```
> [2026-06-05 17:56:58.37 GMT-4] CV Filtered Side B
```

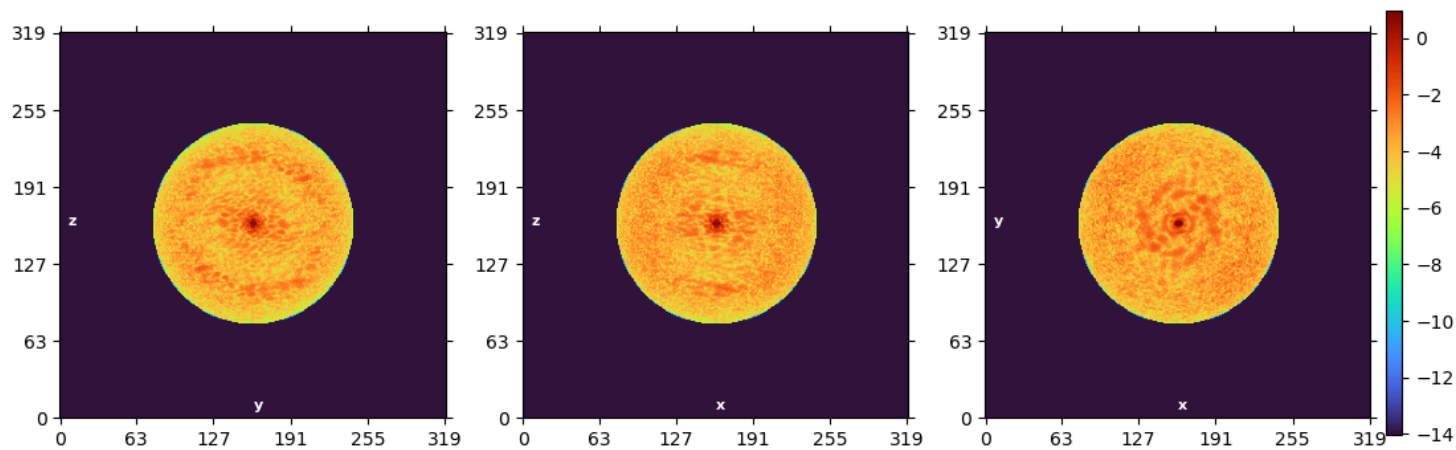


```
> [2026-06-05 17:57:09.38 GMT-4] [CPU: 12.33 GB] [Avail: 287.97 GB] Estimated Bfactor: -54.9
> [2026-06-05 17:57:09.39 GMT-4] [CPU: 12.33 GB] [Avail: 288.06 GB] Plotting..
```

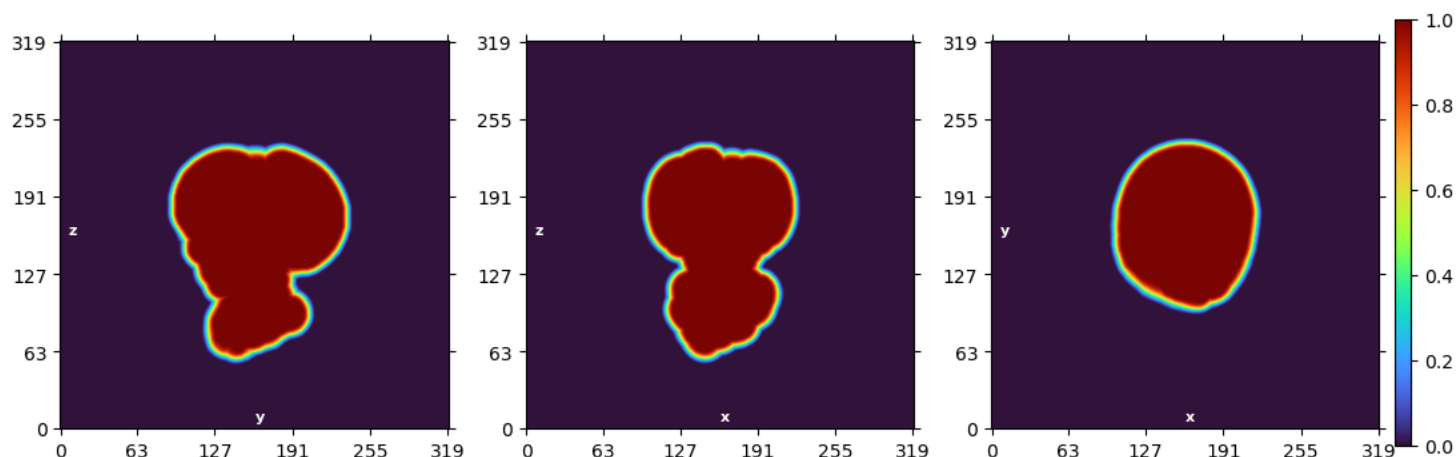
> [2026-06-05 17:57:16.15 GMT-4] Real Space Slices Iteration 005



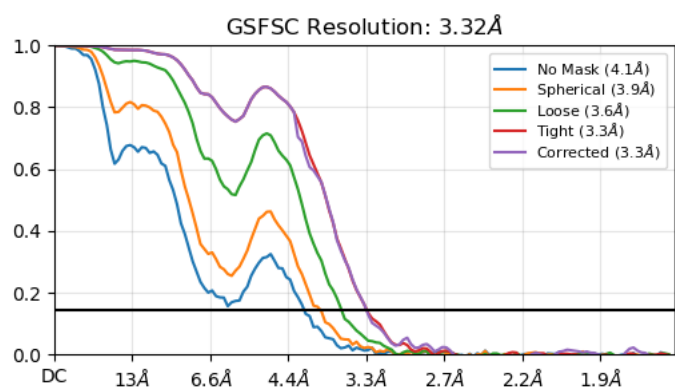
> [2026-06-05 17:57:17.49 GMT-4] Fourier Space Slices Iteration 005



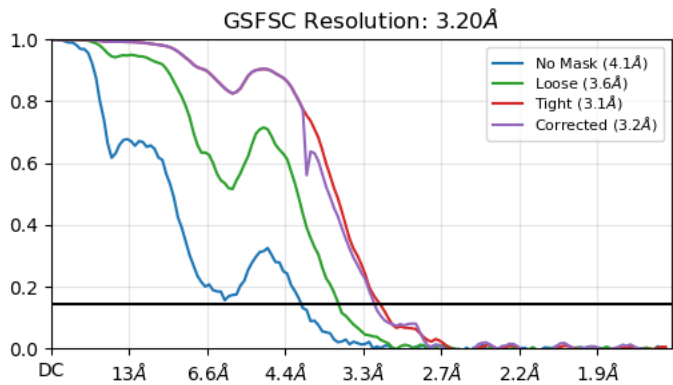
> [2026-06-05 17:57:19.53 GMT-4] Real Space Mask Slices Iteration 005



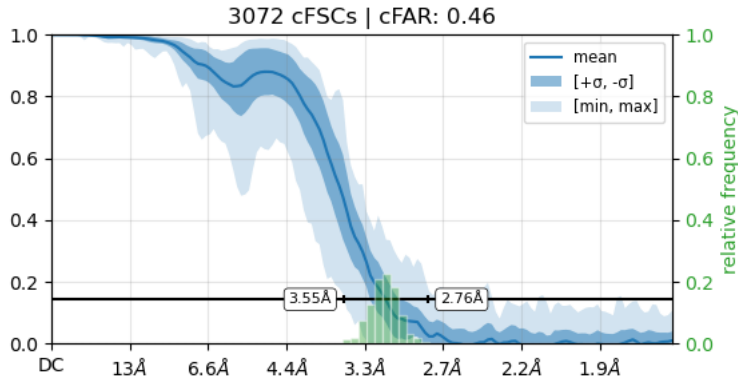
> [2026-06-05 17:57:20.11 GMT-4] FSC Iteration 005



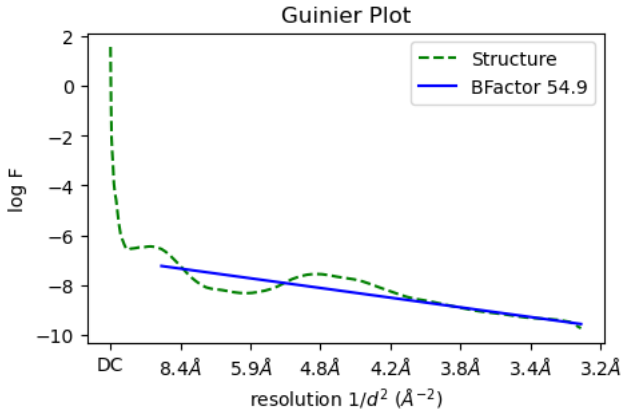
> [2026-06-05 17:57:20.72 GMT-4] FSC Iteration 005, after FSC-mask auto-tightening



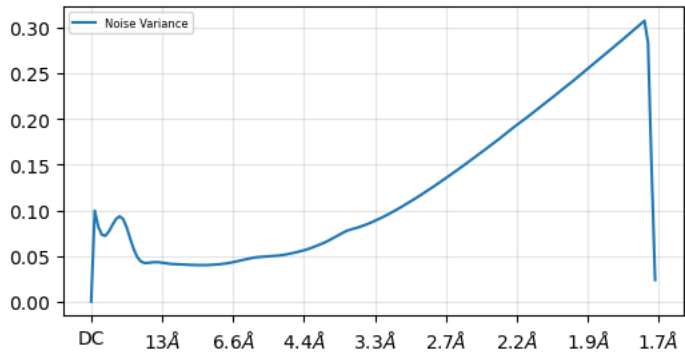
> [2026-06-05 17:57:21.56 GMT-4] cFSCs (Half-angle: 20°) Iteration 005, with autotight mask



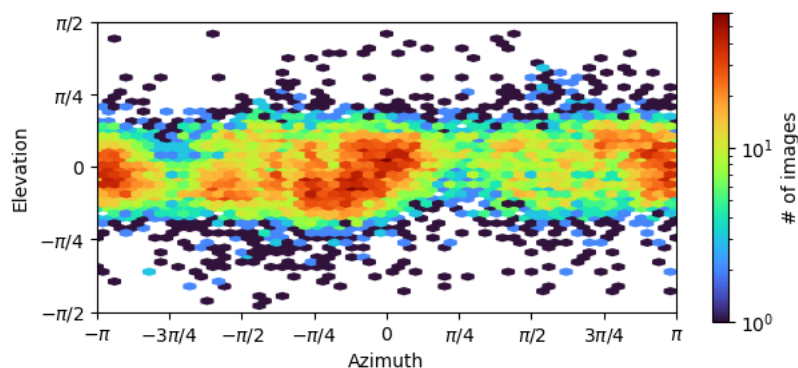
> [2026-06-05 17:57:27.61 GMT-4] Guinier Plot Iteration 005



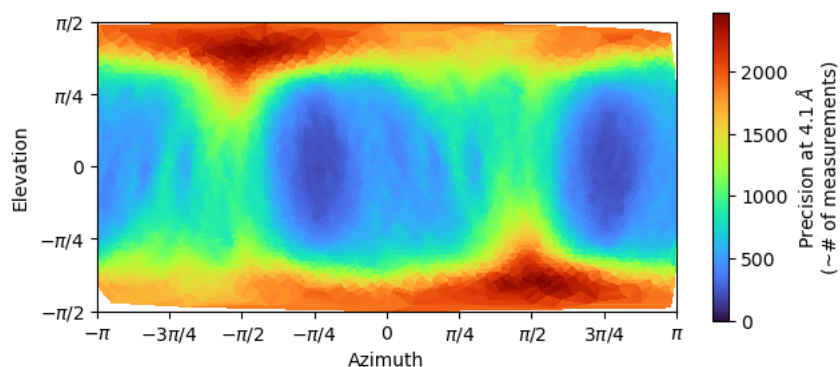
> [2026-06-05 17:57:28.10 GMT-4] Noise Model Iteration 005



> [2026-06-05 17:57:29.09 GMT-4] Viewing Direction Distribution Iteration 005



> [2026-06-05 17:57:30.81 GMT-4] Posterior Precision Directional Distribution Iteration 005



```
> [2026-06-05 17:57:30.82 GMT-4] [CPU: 12.36 GB] [Avail: 289.48 GB] Done in 21.424s.
> [2026-06-05 17:57:30.82 GMT-4] [CPU: 12.36 GB] [Avail: 289.48 GB] Outputting files..
> [2026-06-05 17:57:48.37 GMT-4] [CPU: 12.27 GB] [Avail: 289.98 GB] Done in 17.547s.
> [2026-06-05 17:57:48.37 GMT-4] [CPU: 12.27 GB] [Avail: 289.98 GB] Done iteration 5 in 183.701s. Total time so far 1078.562s
> [2026-06-05 17:57:49.16 GMT-4] [CPU: 12.27 GB] [Avail: 290.76 GB] ===== Done Refinement =====
> [2026-06-05 17:57:49.16 GMT-4] [CPU: 12.27 GB] [Avail: 290.76 GB] Note that the output structure from refinement has been
                                                                    lowpass filtered to the gold-standard FSC resolution
                                                                    estimated
                                                                    above. However, the structure probably needs sharpening in
                                                                    order to best visualize high resolution features. This can be
                                                                    done with the Sharpening task (as a new experiment).

> [2026-06-05 17:57:49.16 GMT-4] [CPU: 12.27 GB] [Avail: 290.79 GB] Full run took 1079.357s
> [2026-06-05 17:57:49.75 GMT-4] [CPU: 6.41 GB] [Avail: 296.96 GB] -----
-

> [2026-06-05 17:57:49.75 GMT-4] [CPU: 6.41 GB] [Avail: 296.97 GB] Compiling job outputs...
> [2026-06-05 17:57:49.76 GMT-4] [CPU: 6.41 GB] [Avail: 296.97 GB] Passing through outputs for output group particles from input
                                                                    group particles

> [2026-06-05 17:57:49.82 GMT-4] [CPU: 6.41 GB] [Avail: 296.97 GB] This job outputted results ['alignments3D', 'ctf']
> [2026-06-05 17:57:49.83 GMT-4] [CPU: 6.41 GB] [Avail: 296.97 GB] Loaded output dset with 13211 items
> [2026-06-05 17:57:49.83 GMT-4] [CPU: 6.41 GB] [Avail: 296.97 GB] Passthrough results ['blob']
> [2026-06-05 17:57:49.92 GMT-4] [CPU: 6.41 GB] [Avail: 296.96 GB] Loaded passthrough dset with 13211 items
> [2026-06-05 17:57:49.96 GMT-4] [CPU: 6.41 GB] [Avail: 296.96 GB] Intersection of output and passthrough has 13211 items
> [2026-06-05 17:57:49.99 GMT-4] [CPU: 6.41 GB] [Avail: 296.96 GB] Output dataset contains: ['blob']
> [2026-06-05 17:57:49.99 GMT-4] [CPU: 6.41 GB] [Avail: 296.96 GB] Outputting passthrough result blob
> [2026-06-05 17:57:50.00 GMT-4] [CPU: 6.41 GB] [Avail: 296.96 GB] Checking outputs for output group particles
> [2026-06-05 17:57:50.09 GMT-4] [CPU: 6.41 GB] [Avail: 296.95 GB] Updating job size...
> [2026-06-05 17:57:50.67 GMT-4] [CPU: 6.41 GB] [Avail: 296.92 GB] Exporting job and creating csg files...
> [2026-06-05 17:57:51.36 GMT-4] [CPU: 6.41 GB] [Avail: 296.51 GB] *****
                                                                    **

> [2026-06-05 17:57:51.36 GMT-4] [CPU: 6.41 GB] [Avail: 296.50 GB] Job complete. Total time 1081.58s
```